

Probability, Statistics & Data Analysis

Lecture 9: Multiple Comparisons

**If you think that statistics has
nothing to say about what you do or
how you could do it better, then you
are either wrong or in need of a
more interesting job**

Stephen Senn

Lecture 9: Multiple Comparisons

What we will cover

- Introduction to ANOVA
 - Why not multiple t-tests?
 - Bonferroni and Bonferroni-Holm
 - Sums of Squares
 - Partitioning Variability
 - The F-statistic
- Post hoc tests
- Kruskal-Wallis Test
- Repeated Measures ANOVA
 - Partitioning Variability
- Two-Way ANOVA
 - Partitioning Variability
 - Interactions





Why not just do lots of t -tests?

ANOVA already

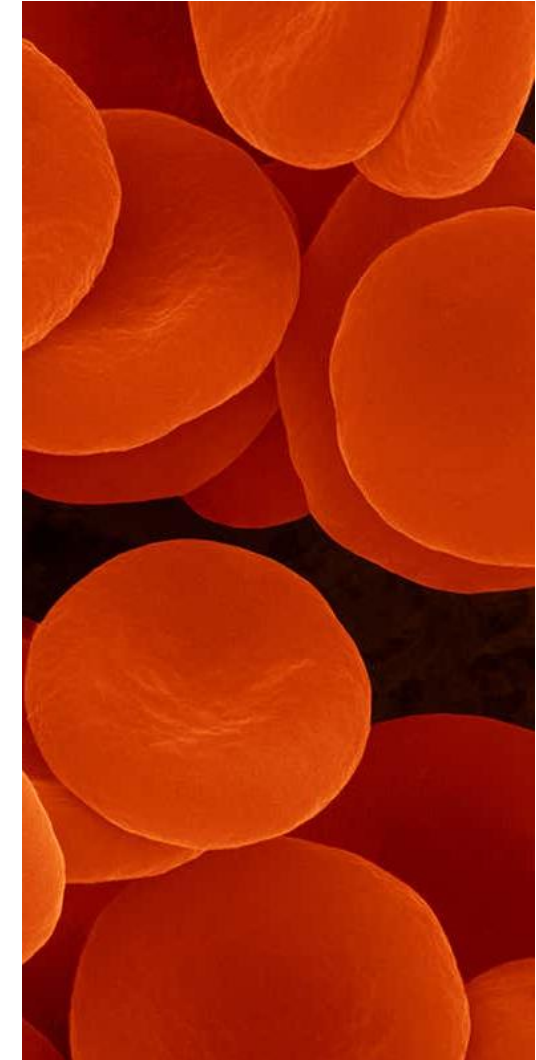
Recapping 1 and 2 sample tests

- We have now covered z -tests and t -tests
 - In practice we tend to use the latter, as the t -test tends towards the z -test for large samples
- The t -test allows us to ask questions about the sample mean from one or two samples
 - One sample – we are interested in a single sample relative to a known value
 - E.g. the outer diameter of 30G needles relative to the standard of 312 μm
 - Two sample – we are interested in the difference between two sample means
 - E.g. the average glucose levels in patient groups receiving an active drug or placebo treatment
- So what do we do if we have more than two samples?
 - E.g. A comparison of **4** different manufacturing techniques for 30G needles
 - E.g. a comparison of glucose levels in patients with placebo **vs** standard insulin therapy **vs** a new automated insulin injector

RBC Deformability



- RBC properties affect fluid dynamics, interactions with endothelial cells and hence the pathophysiology of microvascular diseases
- One key property of RBCs is their deformability. This decreases in diseases such as diabetes, malaria and sickle cell anaemia
- To carry out controlled investigations, we can take blood samples and modify them chemically to alter the deformability
- Let us consider four levels of the factor: treatment
 1. Normal, healthy cells (no treatment)
 2. Diamide treatment (hardens the membrane only)
 3. Glutaraldehyde treatment at 0.05% (hardens both membrane and cytoplasm)
 4. Stored RBCs (as in transfusion – RBC volume increases and membrane stiffens)
- Consider that the observations in each of these factors is independent

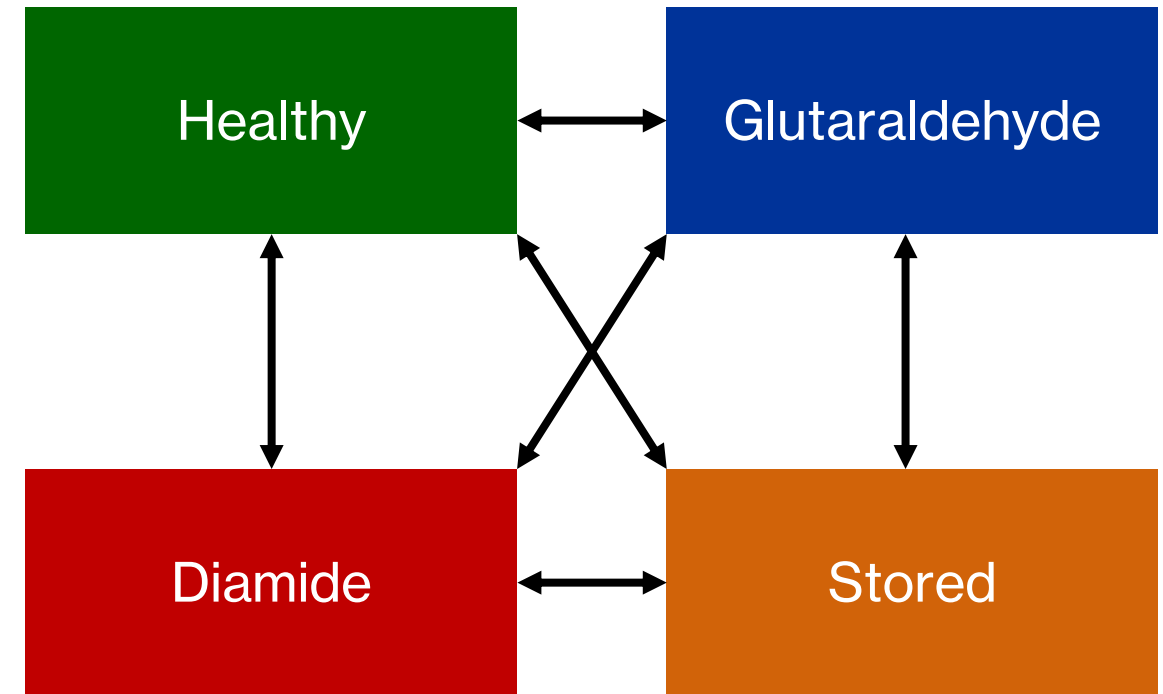


images.newscientist.com/wp-content/

Multiple *t*-tests

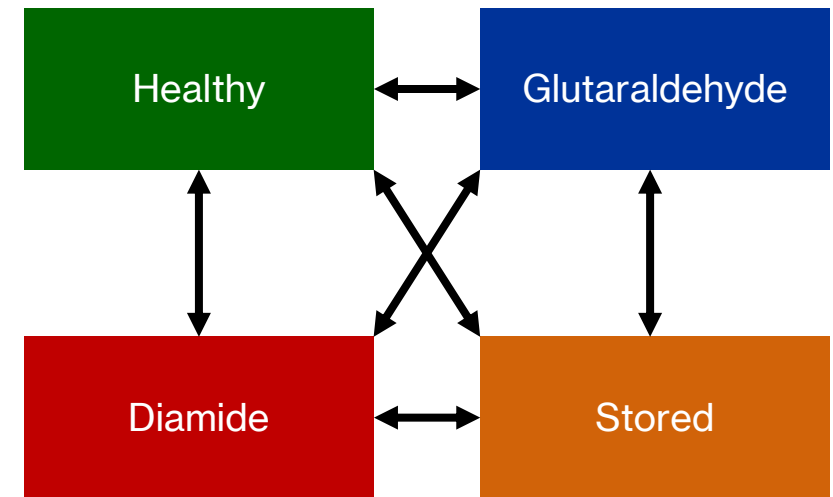
- We take cells from a single donor, separate them into four groups and treat them with
 - Nothing (healthy)
 - Glutaraldehyde
 - Diamide
 - Compare to stored cells from 1 month ago from the same donor
- We measure a 'deformability index' under controlled stress, e.g. using a microchannel
- We measure 10 cells per sample/group
- To compare we start by using *t*-tests with $\alpha = 0.05$

Faivre et al. (2020)



Multiple *t*-tests

- For a single test with $\alpha = 0.05$
 - The probability of getting a Type I error (FP) is $p = 0.05$
 - The chance of not getting a Type I error (given H_0) is 0.95
- For 6 tests
 - The probability of not getting a Type I error (given H_0) $0.95^6 = 0.74$
 - The chance of getting one (or more) Type I errors is $1 - 0.74 = 0.26$
 - I.e. ~ 5 times greater than for a single test
- Our “apparent α ” is rather larger than our intended α
- We trust a ‘positive’ result less than normal
 - because there is a one in four chance it is false

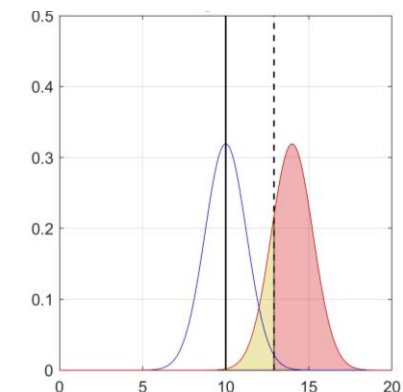
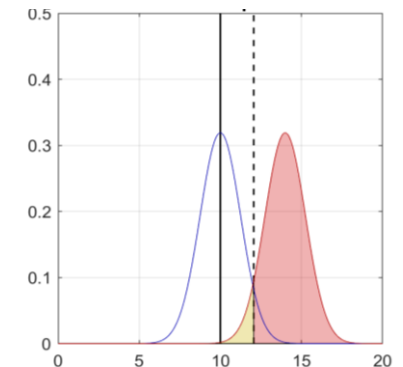
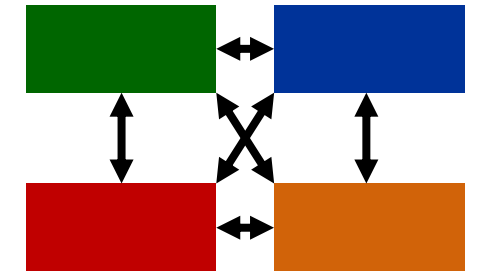


The Bonferroni Correction

- An easy solution is to modify alpha
- For I comparisons (6 in this case) and $\alpha = 0.05$

$$\alpha_B = \frac{\alpha}{I} = \frac{0.05}{6} = 0.0083$$
- Our likelihood of a Type 1 error is then back to our intended $\alpha = 0.05$
 - $(1 - 0.0083)^6 = 0.9512$
 - We compare our p-values to α_B
 - But decreasing α_B increases β
- More intuitively, calculate ‘corrected p-values’

$$p_C = p \times I$$
 - Note that when $p_C > 1$, we treat as $p_C = 1$
- Then compare p_C to $\alpha = 0.05$ (or equivalent)
- However, it is overly ‘conservative’ – lower power



qualitative visualisation

Bonferroni-Holm Method

- Addresses the power issues with the Bonferroni correction
 - Order p values
 - For each, do Bonferroni correction corresponding to the remaining number of comparisons

Remaining comparisons, l	6	5	4	3	2	1
p	0.001	0.004	0.01	0.04	0.1	0.2
$p_C = 6p$ (B)	0.006	0.024	0.06	0.24	0.6	1.2
$p_C = p l$ (B-H)	0.006	0.02	0.04	0.12	0.2	0.2

- If H_0 is true
 - BH treats smallest p-value the same as B
 - Hence similar false positive rate
- If H_0 is false
 - BH treats highest p-value less conservatively
 - Hence lower false negative rate: more powerful

Simulated Multiple Comparisons I



See [Lecture9_code.m](#)

- Consider a comparison of 3 groups
- 1000 cases where H_0 is true
 - Expected proportion of FP is 14.3% ($1 - 0.95^3 = 0.143$)
 - Calculate % FP (different each time)
 - Without adjustment **12.2%** (different each time)
 - Bonferroni **4.4%**
 - Bonferroni-Holm **4.4%**
 - ANOVA **4.5%**
- 1000 cases where H_0 is false
 - Proportion of TP depends on effect size and variability
 - Calculate power to detect all 3 differences (different each time)
 - Without adjustment **81.7%**
 - Bonferroni **61.6%**
 - Bonferroni-Holm **81.4%**
 - ANOVA **100%** (for these specific parameters and sample size)

$$\mu = [2.5 \ 2.5 \ 2.5]$$

$$\sigma = 0.4$$

$$n_{obs} = 15$$

$$\mu = [3.5 \ 3.0 \ 2.5]$$

$$\sigma = 0.4$$

$$n_{obs} = 15$$



Introduction to One-Way ANOVA

More than 2 groups, as defined by one factor

ANOVA overview

- Bonferroni-Holm increases power, but it is still low
- We turn to **Analysis Of VA**riance
 - Compares variance within groups to variance between groups
 - Identifies if there are differences between any groups – *omnibus* test
 - If ANOVA identifies differences, '*post hoc*' tests are required to identify which groups are different from each other
 - Like smarter versions of Bonferroni and Bonferroni-Holm
- Single Factor or One-way ANOVA
 - **Factor** – the characteristic that differentiates treatments or populations
 - Also known as independent variable /explanatory variable /predictor variable
 - E.g. material for artificial hip joint
 - '**Level**' of a factor (I prefer the terminology '**group**')
 - E.g. metal, plastic, ceramic
- Two Factor or Two-way ANOVA
 - Two characteristics that differentiate the treatments/populations
 - E.g. material for head and material for socket



One-way ANOVA notation

- The aim is to infer whether there are any differences in population means between I groups
 - I the number of levels (groups)
 - J_i the number of observations for a given level (group)
 - If $J_1 = J_2 = \dots = J_I = J$ then 'balanced', otherwise 'unbalanced'
 - $\mu_1, \mu_2, \dots, \mu_I$ the populations means of the groups
 - $N = \sum_{i=1}^I J_i$ the total number of observations (general)
- X_{ij} is a random variable that denotes the j^{th} the observation from the i^{th} group
- x_{ij} is the observed value of $X_{i,j}$ after the measurement has been performed
- Assumptions:
 - **all** I groups are sampled from normal distributions with similar variance

ANOVA general concept

- Group means are $\mu_i = \mu + \delta_i$
- The hypotheses are
 - H0: $\mu_1 = \mu_2 = \dots = \mu_I = \mu$ $\delta_i = 0$
 - H1: at least two of the μ_i are different $\delta_i \neq 0$
- Like with regression, ANOVA works to determine which is the best model of the data, either:
 - The grand mean of all the data (H₀) $X_{ij} = \bar{X} + \varepsilon_{ij}$
 - Individual means for each group (H₁) $X_{ij} = \bar{X}_i + \varepsilon_{ij}$
- We calculate sum of squares for
 - ε_{ij} under H₀
 - The total variation from the grand mean, (T - total)
 - ε_{ij} under H₁
 - The residual variations from group means, (R - residual)
 - The model compared to the grand mean
 - The differences between groups means and the grand mean (M - model)

$$outcome_{ij} = (model) + error_{ij}$$

$$X_{ij} = \mu + \delta_i + \varepsilon_{ij}$$

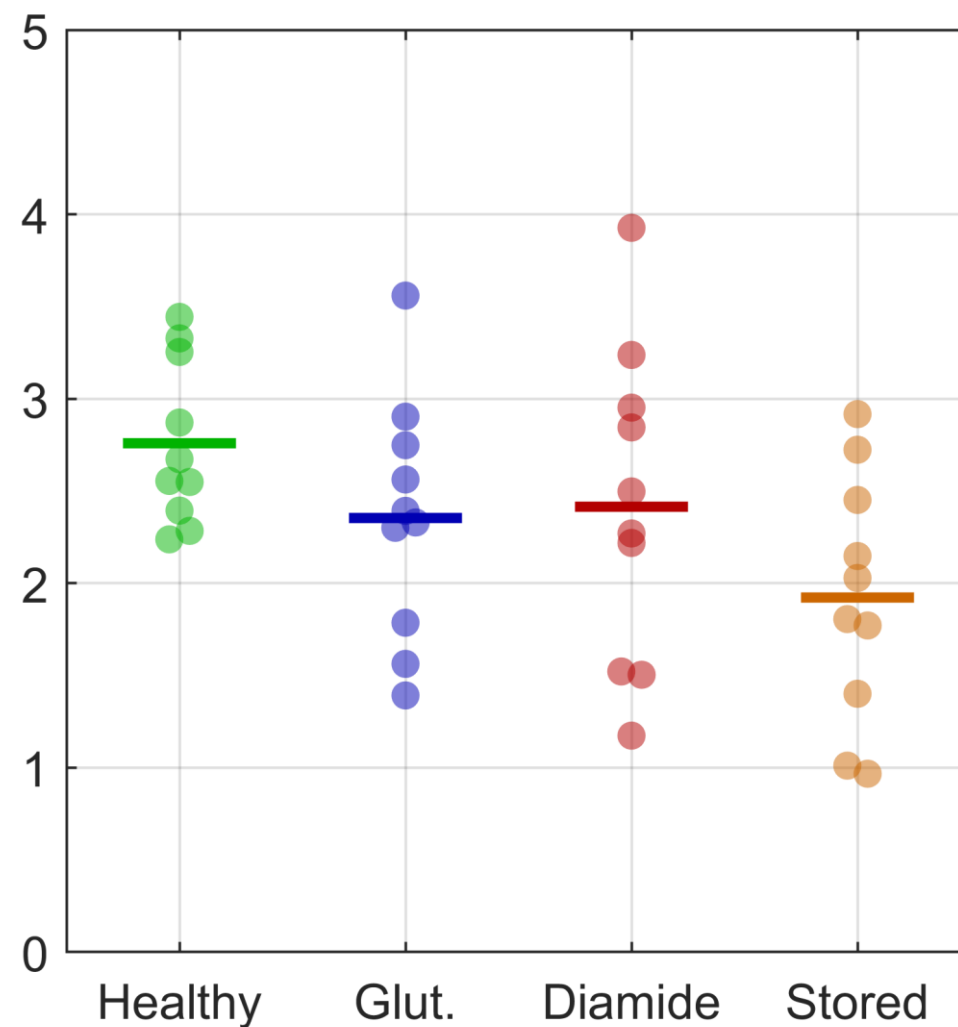
- μ is the 'grand mean' ($\bar{x} = \hat{\mu}$)
- δ_i is the 'treatment effect' ($\bar{d}_i = \hat{\delta}_i$)
- $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ random deviations from our model
- This is my notation for consistency with d from paired tests
- In terms of observed values:

$$X_{ij} = \bar{X} + \bar{D}_i + \varepsilon_{ij} = \bar{X}_i + \varepsilon_{ij}$$



Data for RBC deformability

- 1 Factor: the type of treatment
- $I = 4$ levels (groups)
- $J = 10$ observations per group (balanced)
- $N = IJ = 40$ observations in total





Defining Sample Means

	Healthy <i>i</i> = 1	Glut. <i>i</i> = 2	Diamide <i>i</i> = 3	Stored <i>i</i> = 4	
<i>j</i> = 1	3.42	3.15	2.35	1.59	
<i>j</i> = 2	2.79	2.94	0.78	3.55	
<i>j</i> = 3	3.48	1.66	1.61	1.24	
<i>j</i> = 4	3.48	2.82	2.89	2.26	
<i>j</i> = 5	1.25	1.90	2.36	0.88	
<i>j</i> = 6	1.94	3.71	1.51	1.52	
<i>j</i> = 7	1.85	2.47	1.90	1.63	
<i>j</i> = 8	3.32	3.81	2.60	2.42	
<i>j</i> = 9	4.18	2.16	1.61	2.85	
<i>j</i> = 10	2.74	2.97	3.18	2.34	
$\bar{x}_i$	2.84	2.76	2.08	2.03	2.43

Sample means

$$\bar{x}_i = \frac{1}{J} \sum_{j=1}^J x_{i,j}$$

Grand mean

$$\bar{x} = \frac{1}{N} \sum_{k=1}^N x_k = \frac{1}{N} \sum_{i=1}^I \sum_{j=1}^{J_i} x_{ij}$$

$$= \frac{1}{I} \sum_{i=1}^I \bar{x}_i$$

$\bar{x}$



Sums of Squares

Quantifying Sources of Variance



Total variation from the grand mean: H_0

Total Sum of Squares

Deviations from the grand mean

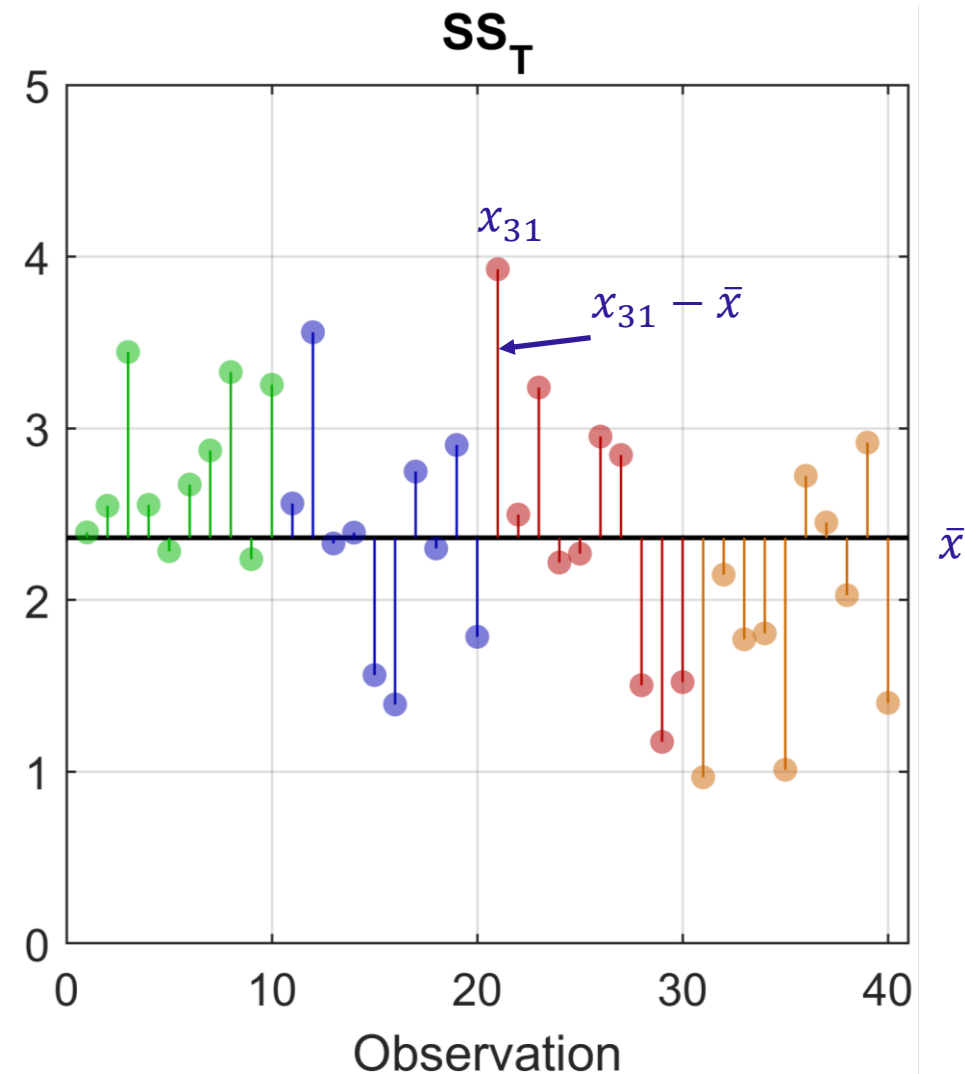
$$SS_T = \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X})^2$$

$$\nu_T = N - 1$$

Alternatively

$$s_{grand}^2 = \frac{1}{N-1} \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X})^2$$

$$SS_T = (N-1)s_{grand}^2$$





Residual variation from the model: H_1

Residual Sum of Squares

Deviations from the model of individual means (ε_{ij})

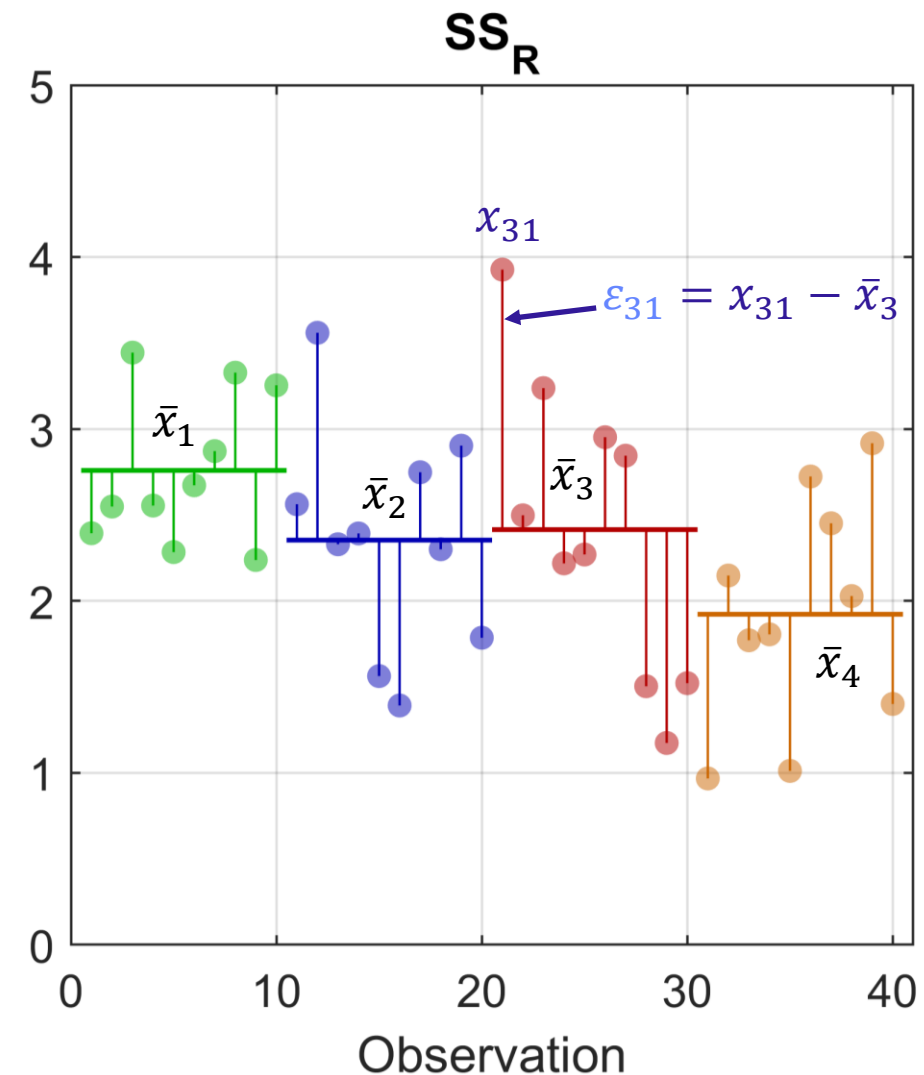
$$SS_R = \sum_{i=1}^I \sum_{j=1}^{J_i} (X_{ij} - \bar{X}_i)^2$$

1 mean per group

$$\nu_R = N - I$$

Alternatively

$$SS_R = \sum_{i=1}^I s_i^2 (J_i - 1)$$





Model variation from the grand mean

Model Sum of Squares

Deviations of the individual means from the grand mean ($\bar{D}_i$)

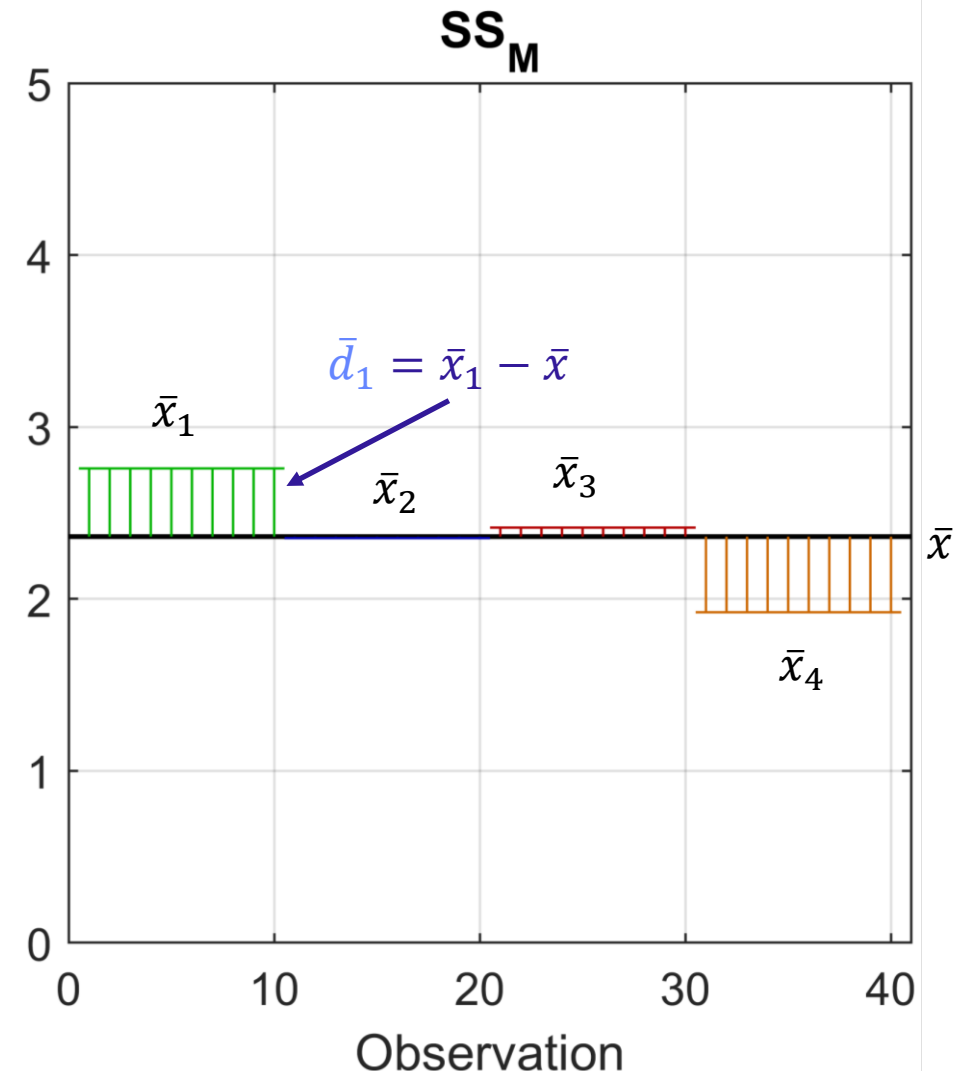
$$SS_M = \sum_{i=1}^I J_i (\bar{X}_i - \bar{X})^2$$

1 mean per group

1 grand mean

$$v_M = I - 1$$

J_i weights the contribution by the sample size for group I

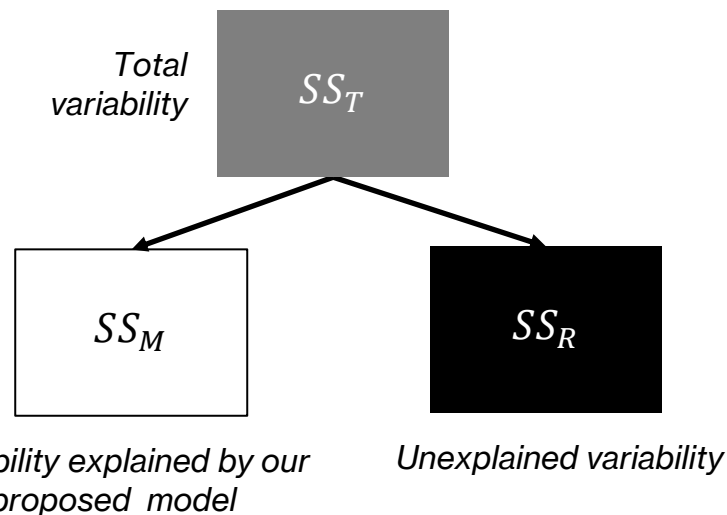




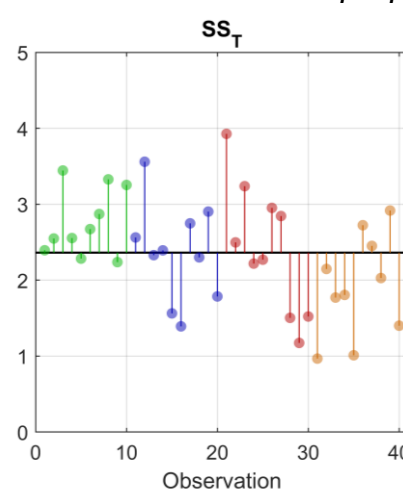
Partitioning Variability

$$X_{ij} - \bar{X} = \bar{D}_i + \varepsilon_{ij}$$

- How do our sums of squares relate to each other?
 - Alternatively, what are the sources of the variability from the null model $X_{ij} - \bar{X}$
 - We can see visually that
- $$X_{ij} - \bar{X} = (\bar{X}_i - \bar{X}) + (X_{ij} - \bar{X}_i)$$
- We don't actually need all three Sum Squares
 - Which 2 do we want?

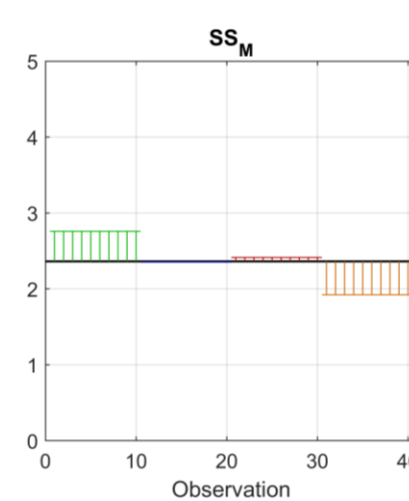


$$SS_T = SS_R + SS_M$$



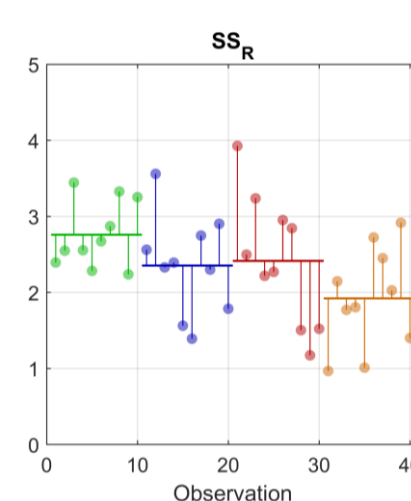
$SS_T: X_{ij} - \bar{X}$
Total variability

=



$SS_M: \bar{D}_i$
Variability explained by our proposed model

+



$SS_R: \varepsilon_{ij}$
Unexplained variability

The test-statistic and mean squares

$$\text{Test statistic} = \frac{\text{Effect}}{\text{Error}} = \frac{\text{Variance explained by model}}{\text{Variance not explained by the model}}$$

From SS_M From SS_R

- SS_M – how much of the variability can be explained by the model of individual group means rather than the grand mean
 - depends on I values (a mean for each group), relative to 1 grand mean
$$v_M = I - 1$$
- SS_R – remaining variation (random noise)
 - Depends on $N = I \times J$ values (a value for each data point), relative to I group means
$$v_R = N - I$$
- Convert to Mean Square values (independent of number of observations) using DOF
- We don't need SS_T directly, but sometimes helpful for calculations

$$v_T = N - 1 = v_R + v_M$$
- Note that for regression, s_ε^2 is the MS_R but there are different DOF

$$MS_M = \frac{SS_M}{v_M} = \frac{SS_M}{I - 1}$$

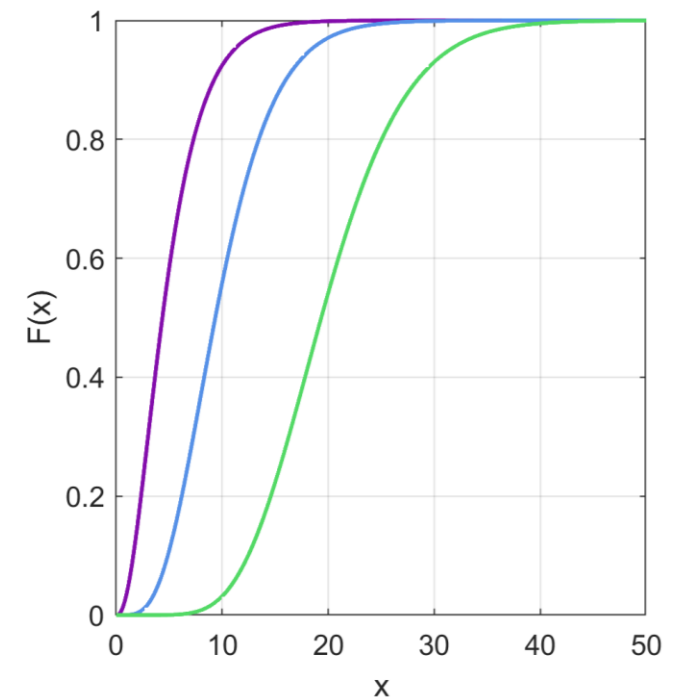
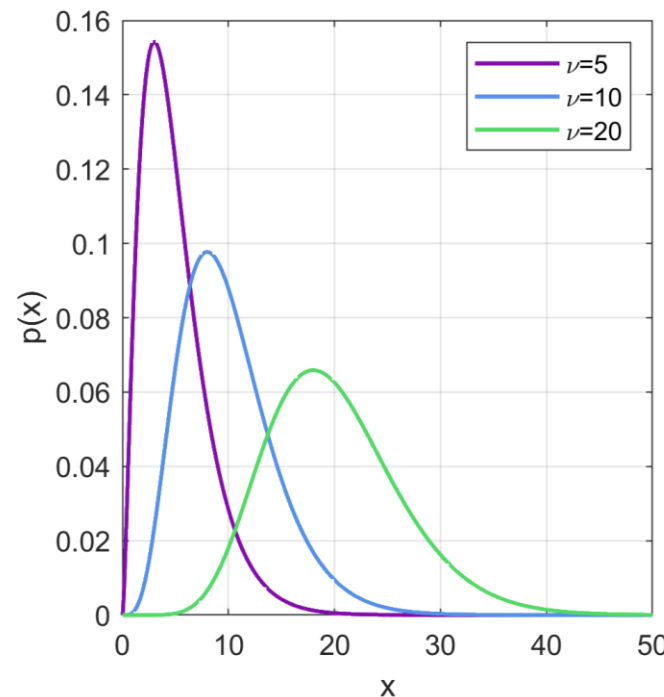
$$MS_R = \frac{SS_R}{v_R} = \frac{SS_R}{N - I}$$

Note I is sometimes notated as k

FYI - χ^2 distribution

- The Mean Squares values follow a χ^2 distribution
- Don't worry about the PDF, other than to note it is dependent on DOF
- Use tables to find critical values of $\chi^2_{\alpha, \nu}$
- Or use `chi2inv` and `chi2cdf` as required

$$f(x) = \frac{x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{\nu}{2}} \Gamma\left(\frac{\nu}{2}\right)}, x \geq 0$$





The F-statistic

and its distribution

The F-statistic

- For single factor ANOVA the test statistic is therefore

$$F = \frac{MS_M}{MS_R}$$

Effect
Error

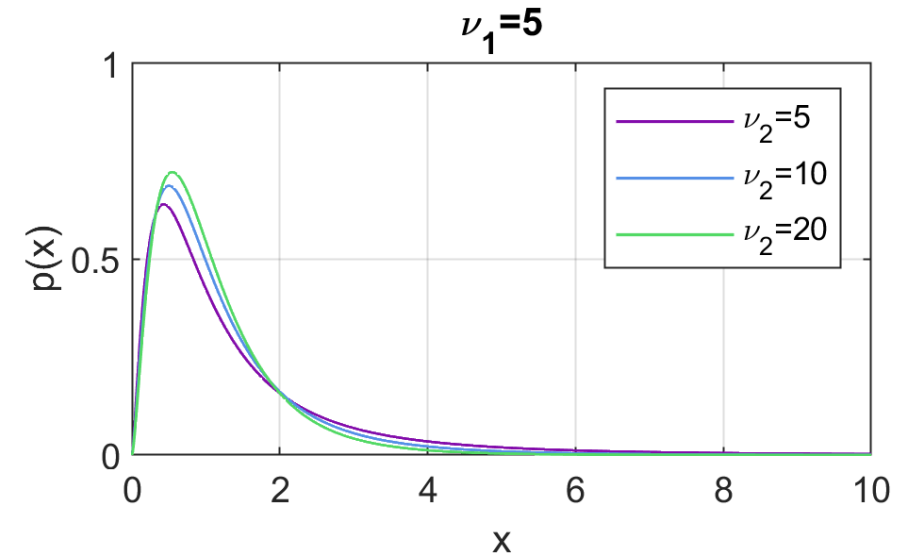
Variation explained by model
Variation from random noise

Between groups
Within groups

- The F statistic has an F distribution
 - $\nu_1 = \nu_M = I - 1$
 - $\nu_2 = \nu_R = N - I$
- It is a ratio of two χ^2 distributions

$$F = \frac{\chi_1^2 / \nu_1}{\chi_2^2 / \nu_2}$$

- Use tables to find $F_{crit}(\alpha, \nu_1, \nu_2)$ Or use `finv` and `fcdf` as required



Using the F statistic

- Select a value of α (in advance) and hence a rejection region:

$$f \geq f_{crit} = f_{\alpha, \nu_M, \nu_R} = f_{\alpha, \nu_1, \nu_2}$$

- Note, always $\geq$ as tailedness is not in question
 - Hence not on reference sheet
- Check if f is in the rejection region
 - If yes, reject H_0
 - If no, do not reject H_0

- Alternatively, calculate p-value (F is CDF for f)

$$p = 1 - F(f, \nu_M, \nu_R)$$

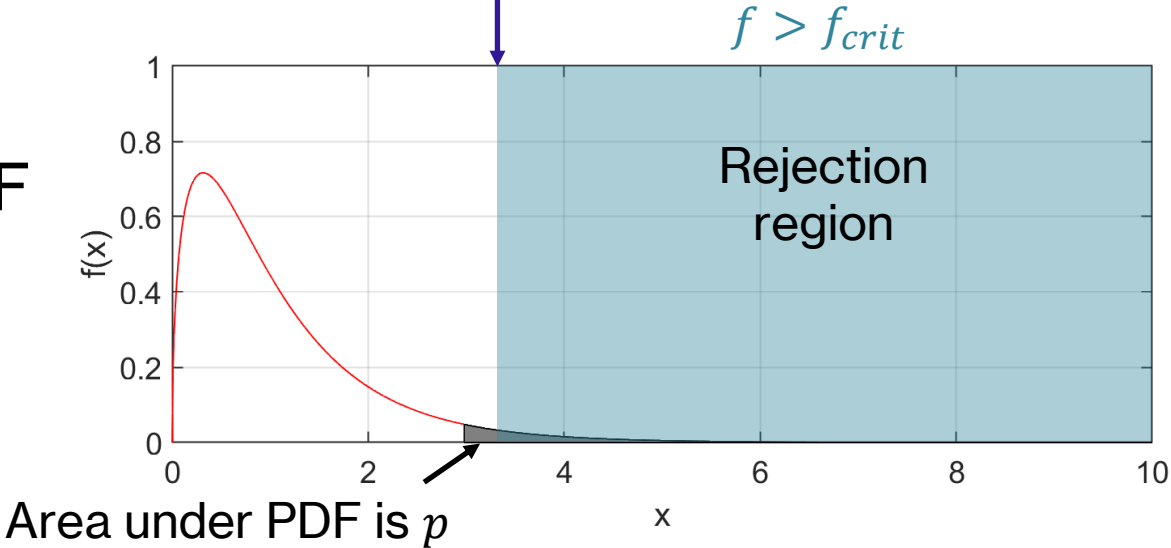
Table 3: Selected Critical values of the F-distribution



ν_1

e.g. $\nu_1 = 5, \nu_2 = 10$

		ν_1														
		1	2	3	4	5	6	7	8	9	10	12	15			
ν_2	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86			
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62			
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94			
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51			
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22			
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01			
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85			
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72			
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62			
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53			
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46			



Rabbit Carotids



- Langille and O'Donnell (1986) wanted to investigate the hypothesis that changes in shear stress result in vascular remodelling, mediated by endothelial cells
- The factor (independent variable) was the group, characterised by
 - Control (no pre-treatment)
 - SM (Smooth Muscle cells inactivated with a muscle relaxant)
 - EC (Endothelial Cells chemically denuded prior to the ligation)

$I = 3$ groups
- Dependent variable is the diameter of the left common carotid, measured by casting, two weeks after the ligation
- Measured 5 animals in each group

$J = 5$ observations per sample, balanced
- Use $\alpha = 0.05$

Simulated Rabbit Data



The hypotheses are

- H_0 = There is no difference in the population mean carotid diameter for any of the groups
- H_1 = at least two of the population average carotid diameters are different

‘Real-world’ hypothesis

- We expect to see group 3 lower than the other groups

Try not to let this bias your analysis!

	Control $i = 1$	SM $i = 2$	EC $i = 3$	
$j = 1$	3.1	2.5	2.2	
$j = 2$	3.2	2.6	2.0	
$j = 3$	3.2	3.0	2.5	
$j = 4$	2.5	3.6	1.7	
$j = 5$	2.8	2.7	2.1	$\bar{x}$
$\bar{x}_i$	2.96	2.88	2.08	2.66
s_i^2	0.093	0.197	0.113	

Sample means Sample variances Grand Mean

Rabbits - Sum Squares



$$SS_M = \sum_{i=1}^I J_i (\bar{X}_i - \bar{X})^2$$

$$\nu_M = I - 1$$

$$SS_R = \sum_{i=1}^I \sum_{j=1}^J (X_{ij} - \bar{X}_i)^2 = \sum_{i=1}^I S_i^2 (J_i - 1)$$

$$\nu_R = N - I$$

	Control <i>i</i> = 1	SM <i>i</i> = 2	EC <i>i</i> = 3	
<i>j</i> = 1	3.1	2.5	2.3	
<i>j</i> = 2	3.2	2.6	2.0	
<i>j</i> = 3	3.2	3.0	2.6	
<i>j</i> = 4	2.5	3.6	1.7	
<i>j</i> = 5	2.8	2.7	2.1	$\bar{X}$
$\bar{X}_i$	2.96	2.88	2.14	2.66
S_i^2	0.093	0.197	0.113	

Rabbits - MSS and F



$SS_M = 2.044$

$\nu_M = 2$

$SS_R = 1.612$

$\nu_R = 12$

Table 3: Selected Critical values of the F-distribution



$F_{crit} = f_{1-\alpha,\nu_1,\nu_2} = F^{-1}(1 - \alpha, \nu_1, \nu_2)$

		ν_1											
		1	2	3	4	5	6	7	8	9	10	12	15
ν_2	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46
	15	4.54	3.68	3.28	3.05	2.90	2.79	2.71	2.64	2.59	2.54	2.47	2.40

ANOVA Table

- ANOVA results can be easily summarised in an ANOVA table

	SS	df	MS	F	P(F H ₀)
Columns	SS_M	$\nu_M = I - 1$	SS_M/ν_M	$F = \frac{MS_M}{MS_R}$	p
Error	SS_R	$\nu_R = N - I$	SS_R/ν_R		
Total	SS_T	$\nu_T = N - 1$			

$SS_T = SS_M + SS_R$ $\nu_T = \nu_M + \nu_R$

	SS	df	MS	F	P(F H ₀)
Columns	2.044	2	1.022	7.608	0.0073
Error	1.612	12	0.1343		
Total	3.656	14			

```
>> 1-fcdf(7.608,2,12)

ans =

    0.0073
```



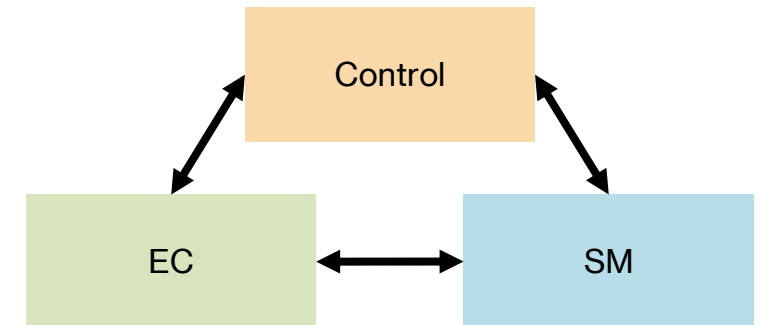


Post hoc testing

There's a difference, but between which groups?

Post hoc tests

- So we have found a difference, what now?
- ANOVA only tells us that at least one of the population means from which the data are sampled is different from at least one of the others
- We can do an additional post hoc (after the event) test to identify pair-wise statistical differences
- The simplest of these are Bonferroni and Bonferroni-Holm, but they are low-power
- For this course, more advanced post-hoc tests are only done in MATLAB (hence not examined)
- See bonus material for some examples

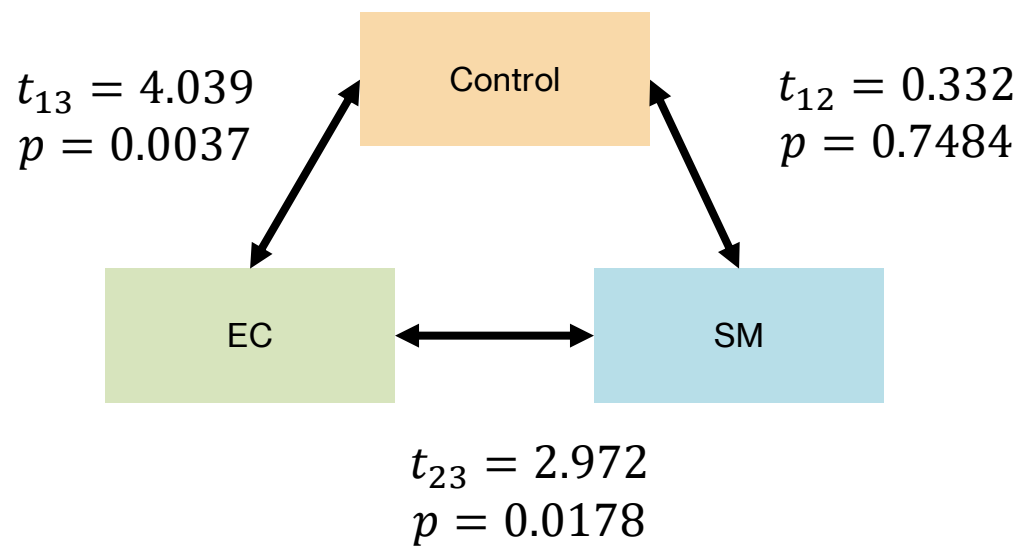


Post hoc testing: Bonferroni-Holm



Hence, we conclude that the average diameter of

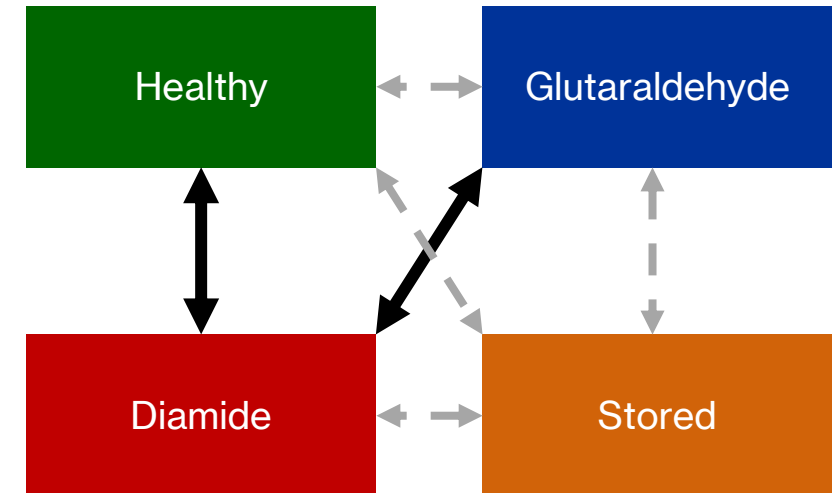
- 1. the EC and Control groups was statistically different
- 2. the EC and SM groups was statistically different
- 3. the SM and Control groups were not statistically different



Reporting Results (back to blood)



- Assumptions of normality and equal variance were not violated (SW-test, Levene's test, $p > 0.05$).
 - More details should be provided in the statistics section of methods (or in an appendix) in a project report/paper
- “There was a statistical difference between the deformability index of the four groups as determined by one-way ANOVA ($F(3,25)=4.39$, $p < 0.05$).”
 - ν_1 ν_2
- “A Tukey post-hoc test revealed that the deformability of diamide treated RBCs was statistically lower than healthy and glutaraldehyde treated cells ($p < 0.05$)”
 - Or an alternative post-hoc test. See Bonus Material.
- “No other statistical differences between groups were found.”
- Should also add details on the means, the differences and the confidence intervals etc. – practical difference
 - confidence intervals are actually modified due to multiple comparisons, and are an output of post-hoc tests





Kruskal-Wallis Test

For non-normal and/or heteroscedastic >two sample analyses

Kruskal-Wallis

- Non-parametric alternative to ANOVA
 - Sometimes called ANOVA on the ranks
 - It isn't quite that (test statistic is different)
- A generalisation of the U-test for more than two groups (k)
- H_0 : the I independent random samples all come from populations with the same population median
 - $\tilde{\mu}_1 = \tilde{\mu}_2 = \dots = \tilde{\mu}_I$ the population medians of the groups
- We rank all the observations as before and generate a test statistic,

$$h = -3(N + 1) \frac{12}{N(N+1)} \sum_{i=1}^I \frac{w_i^2}{n_i}$$

- If all $n_i > 5$
 - If H_0 is true, h is approximated a χ^2 distribution with

$$\nu = I - 1$$

k here is equivalent to I in our ANOVA analysis

Kruskal-Wallis Test

$$r_i = \text{rank}\{x_1, \dots, x_I\},$$

$$w_1 = \sum_{i=1}^{n_1} r_{1,i} \dots$$

$$w_I = \sum_{i=1}^{n_I} r_{I,i}$$

$$N = \sum_{i=1}^I n_i$$

$$h = -3(N + 1) \frac{12}{N(N+1)} \sum_{i=1}^I \frac{w_i^2}{n_i}$$



Young's Modulus of Hydrogels in kPa

- Some researchers want to use tissue engineering to generate bone samples
- They use hydrogels to control Young's modulus as cells respond to the stiffness of the surrounding material
- They want to investigate if there is a difference between gels made by different individuals
- Gels were made by three different researchers who all followed the same protocol
- Is there a statistical difference in the stiffness of gels created by researchers 1, 2 and 3?
- Assume that normality is rejected for at least one of the groups

1	77	54	67	74	71	66	
2	60	41	59	65	62	64	52
3	49	52	69	47	56		



Calculating the H-statistic

Rank the observations

Rank	1	2	3	4.5	4.5	6	7	8	9
Value	41	47	49	52	52	54	56	59	60
Rank	10	11	12	13	14	15	16	17	18
Value	62	64	65	66	67	69	71	74	77

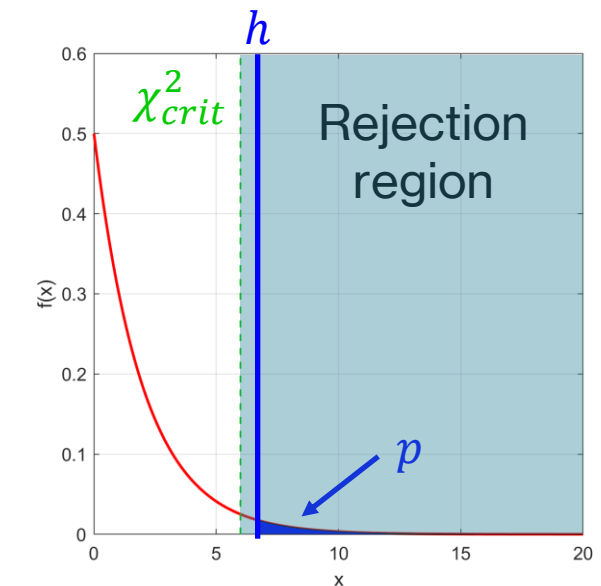
1
2
3

$$h = -3(N + 1) + \frac{12}{N(N + 1)} \sum_{i=1}^I \frac{w_i^2}{n_i}$$

Evaluating Statistical Differences

- We have $h = 6.7$
- We have 3 groups, hence $\nu = I - 1 = 2$
- To evaluate if there is a statistical difference between any of the medians at α
 - Get a critical χ^2_{crit} from the table
 - E.g. for $\alpha = 0.05, \nu = 2, \chi^2_{crit} = 5.99$
 - As $h > \chi^2_{crit}$ we can reject the null hypothesis
- To calculate a p-value
 - $P = 1 - \text{chi2cdf}(h, \nu)$
 - Or could use interpolation
 - For the present data, $p = 0.036$
- We reject the null hypothesis if $\alpha = 0.05$
- We conclude that at least one of the samples comes from a distribution with a different median
- Follow up with post-hoc tests
 - Dunn-Sidak test is recommended

α		0.050	0.010	0.001
		-	-	-
ν	1	3.841	6.635	10.828
	2	5.991	9.210	13.816
	3	7.815	11.345	16.266
	4	9.488	13.277	18.467
	5	11.070	15.086	20.515
	6	12.592	16.812	22.458
	7	14.067	18.475	24.322
	8	15.507	20.090	26.124





Repeated Measures ANOVA

When you use repeats on the same subjects



Repeated Measures Design

- When measurements are not independent
- E.g. for our blood data – we originally assumed that each of the x_{ij} values was independent of all others
 - When independent, variability arises from
 - Variability explained by the model (MS_M)
 - Random deviation from the model (MS_R)
- However, what if each row represented blood from a given individual – a subject
- Each value on that row would no longer be independent - each individual blood sample acts as a ‘baseline’ – not exact, but an RV
- Comparable to ‘paired’ t-tests
- We introduce the average of each row $\bar{X}_j$
- When dependent, variability arises from
 - Variability explained by the model (MS_M)
 - Random deviation from the model ($\bar{X}_i$) (MS_R)
 - Within subjects
 - Differences between individuals ($\bar{X}_j$) (MS_B)
 - Between subjects

$$outcome_{ij} = (model) + error_{ij}$$

$$X_{ij} = \mu + \delta_i + \kappa_j + \varepsilon_{ij}$$

- μ is the ‘grand mean’ ($\bar{x} = \hat{\mu}$)
- δ_i is the ‘treatment effect’ ($\bar{d}_i = \hat{\delta}_i$)
- $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ random deviations from our model
- $\kappa_j \sim N(0, \sigma_\kappa^2)$ random deviations from the average individual
- This is my notation for consistency with d from paired tests
- In terms of observed values:

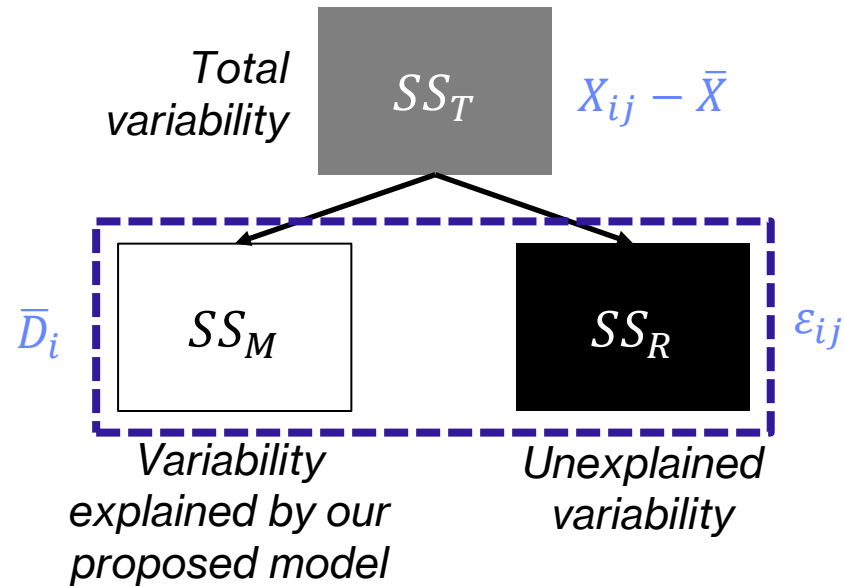
$$X_{ij} = \underbrace{\bar{X} + \bar{D}_i}_{\bar{X}_i} + \bar{K}_j + \varepsilon_{ij}$$

$$\bar{X}_j = \bar{X} + \bar{K}_j$$

Partitioning Variability

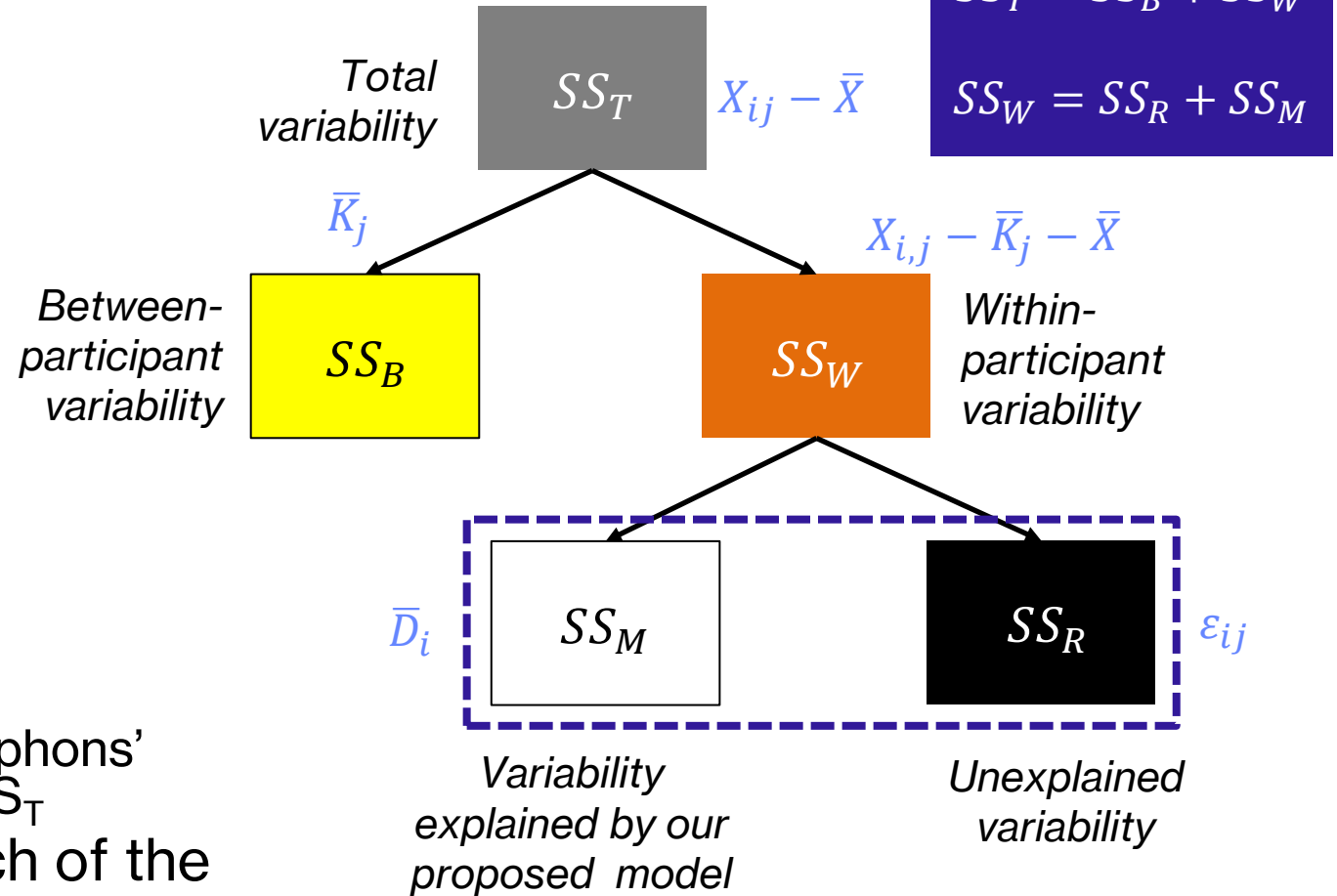
$$X_{ij} - \bar{X} = \bar{D}_i + \varepsilon_{ij}$$

One-Way, independent



$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

Repeated Measures



- For the same total variability
 - Between participant variability 'syphons' some variability, such that $SS_W < SS_T$
- We're still interested in how much of the variability is explained by the group means relative to noise

General Approach

$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

- We still consider

$$\text{Test statistic} = \frac{\text{Effect}}{\text{Error}}$$

- Our model is now that
 - Each group has it's own mean $\bar{X}_i (= \bar{D}_i + \bar{X})$
 - Each individual has their own mean $\bar{X}_j (= \bar{K}_j + \bar{X})$
- When calculating the error (SS_R) we still evaluate the sum squares of deviations from the model
- When calculating effect (SS_M) we are interested in whether the model $\bar{X}_i$ is true (is at least one of $\bar{D}_i$ non-zero)
 - Did the treatments have an effect, irrespective of the variability between individuals
- Our test statistic is still

Subject j	Healthy $i = 1$	Glutaraldehyde $i = 2$	Diamide $i = 3$	Stored $i = 4$	$\bar{X}_j$
$j = 1$	1.38	1.34	1.23	1.15	1.27
$j = 2$	2.45	1.30	1.82	1.44	1.75
$j = 3$	4.93	3.61	2.68	2.58	3.45
$j = 4$	2.62	2.13	2.08	0.97	1.95
$j = 5$	4.04	3.12	2.64	1.67	2.87
$j = 6$	1.72	2.21	1.65	2.62	2.05
$j = 7$	3.73	2.22	2.73	1.71	2.60
$j = 8$	5.20	3.53	3.69	4.03	4.11
$j = 9$	2.83	2.58	1.47	2.28	2.29
$j = 10$	0.67	0.51	0.73	0.87	0.70
$\bar{X}_i$	2.96	2.25	2.07	1.93	2.43

$$F = \frac{MS_M}{MS_R}$$



Model variation from the grand mean

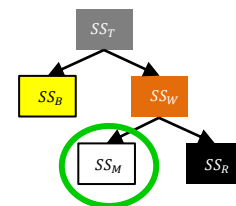
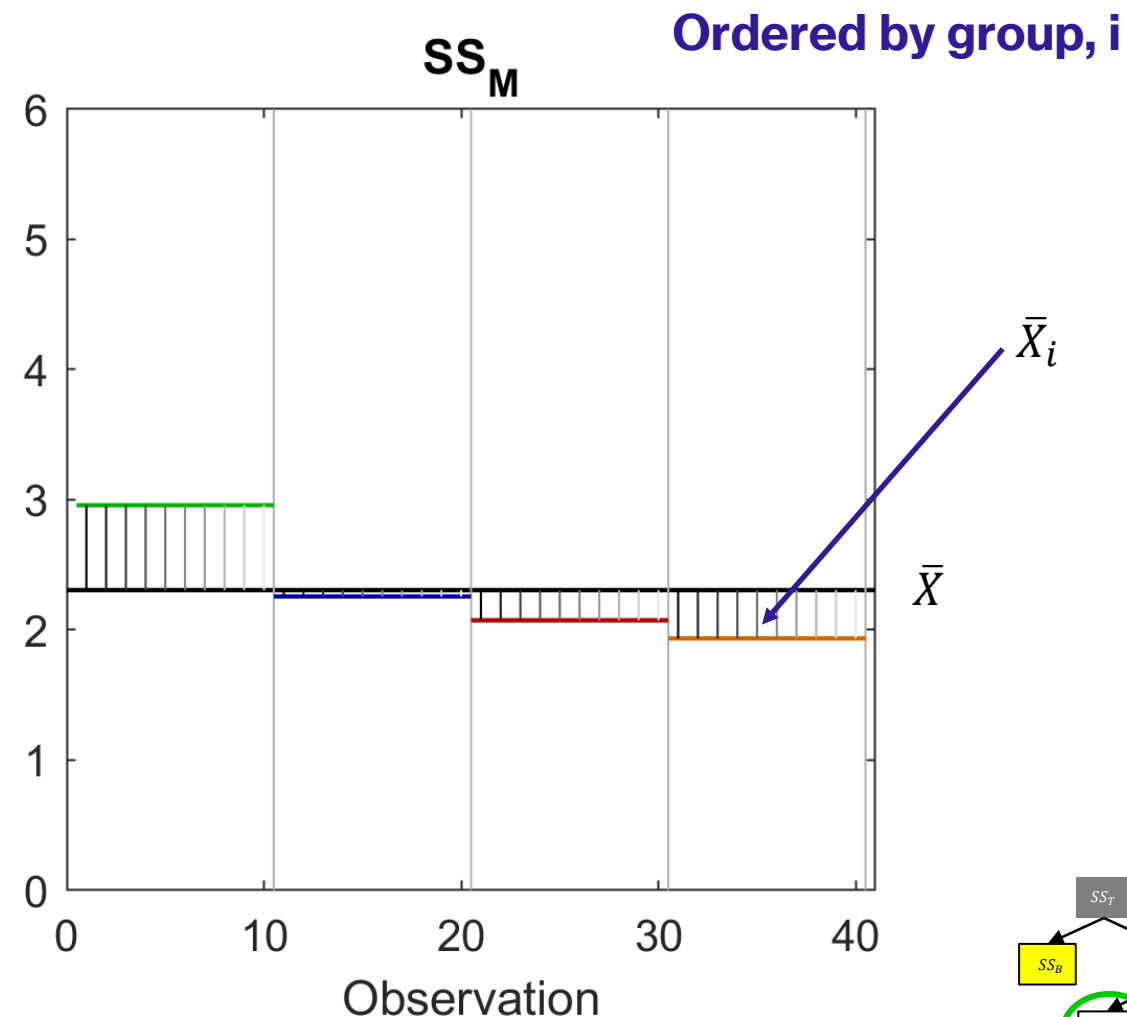
Exactly the same as for independent

Model Sum of Squares

Deviations of the group means
(the part of the model we are
interested in) from the grand
mean

$$SS_M = \sum_{i=1}^I J(\bar{X}_i - \bar{X})^2$$

$$v_M = I - 1$$

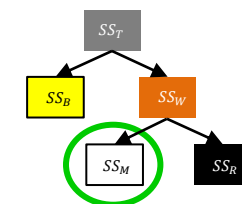
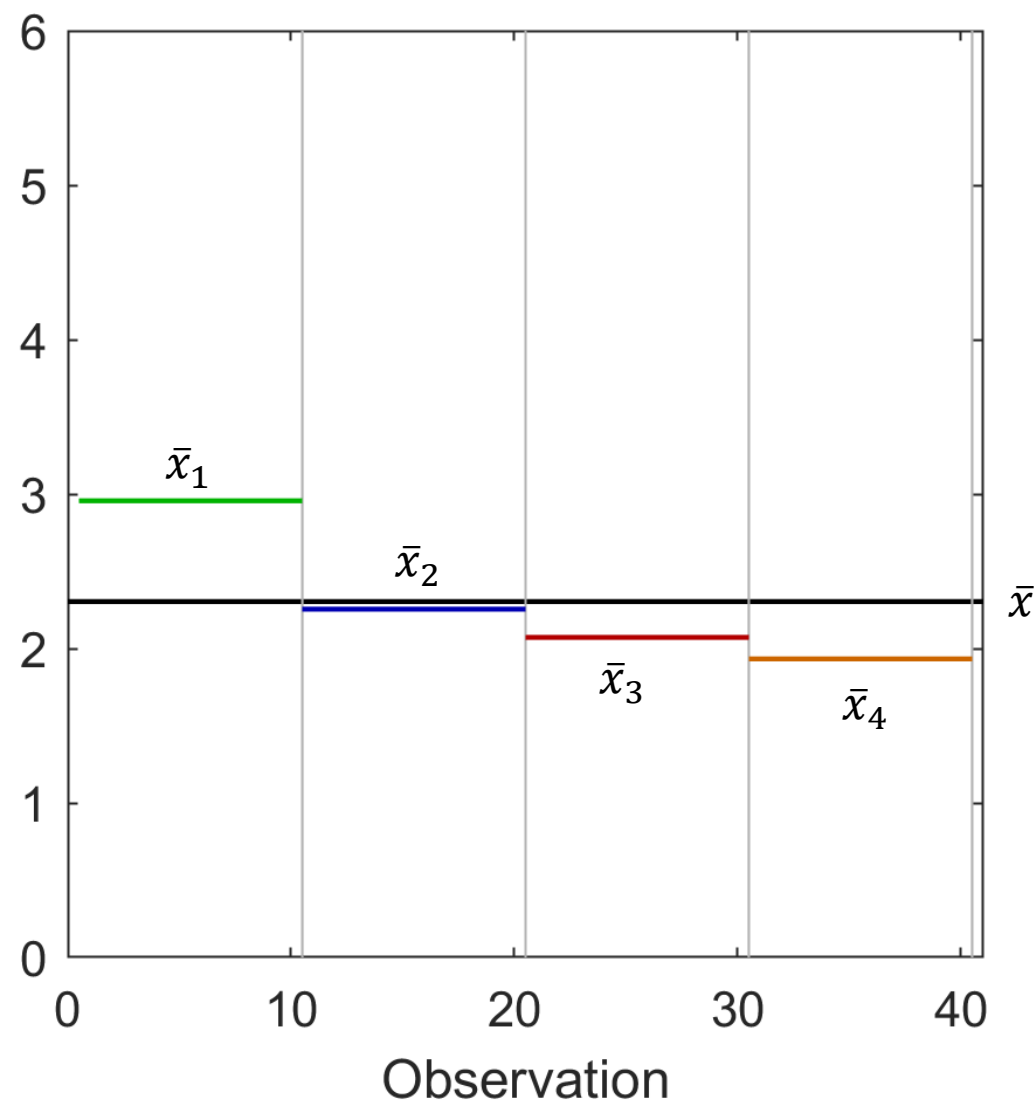


Residual variation from the model



$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

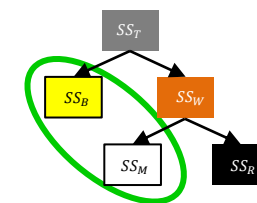
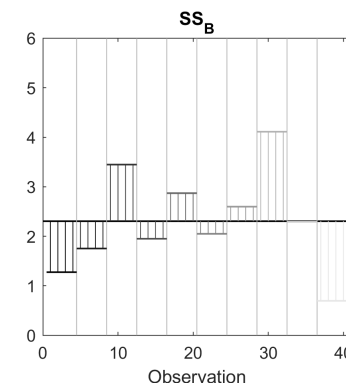
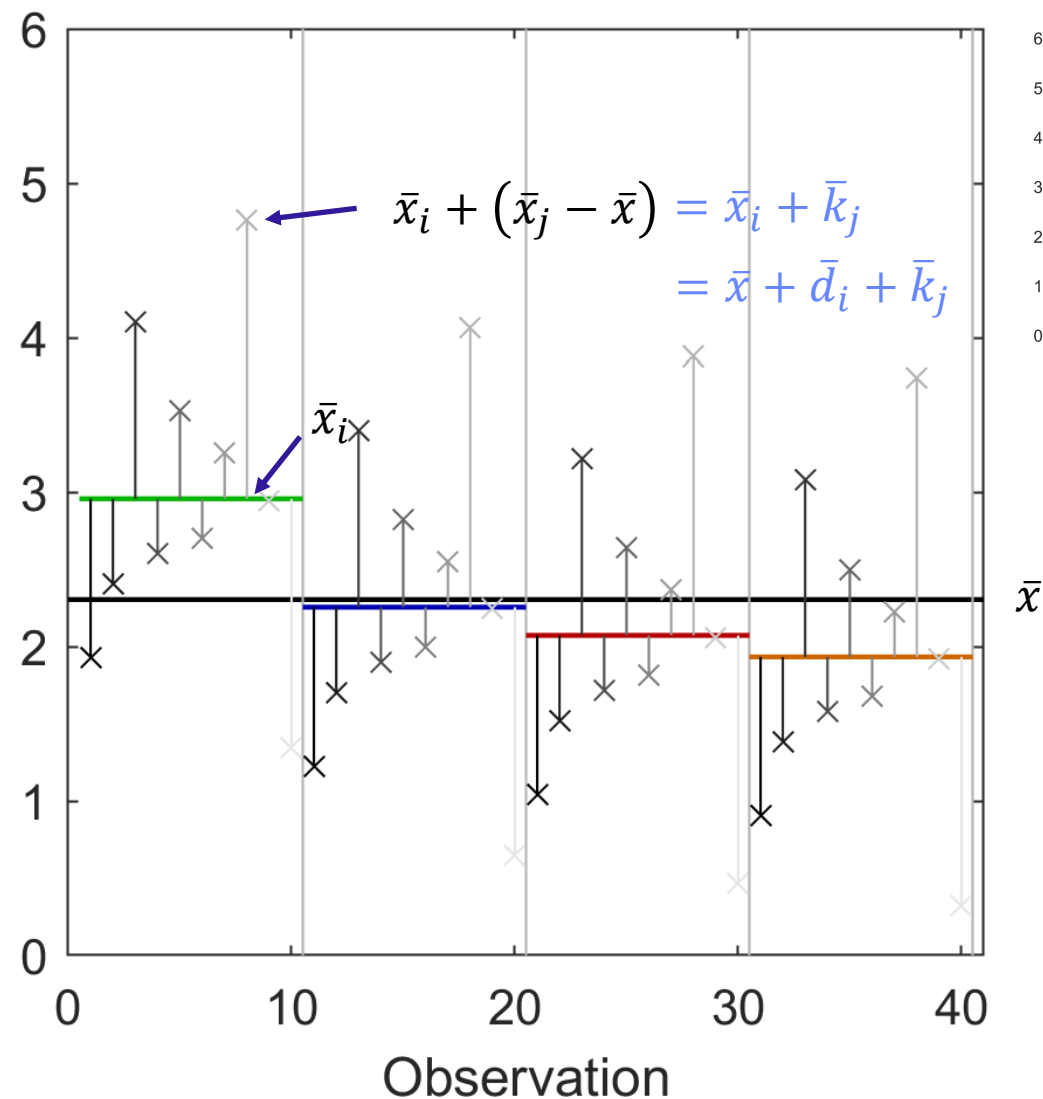
- Each group has a different group mean $\bar{X}_i$



Residual variation from the model

$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

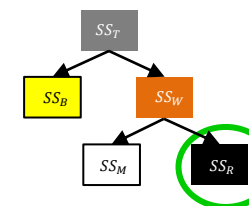
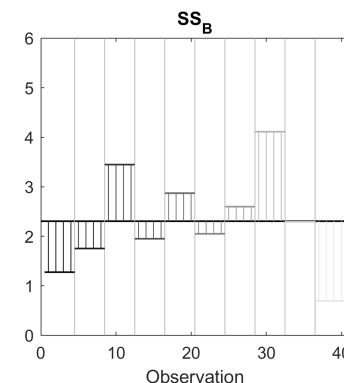
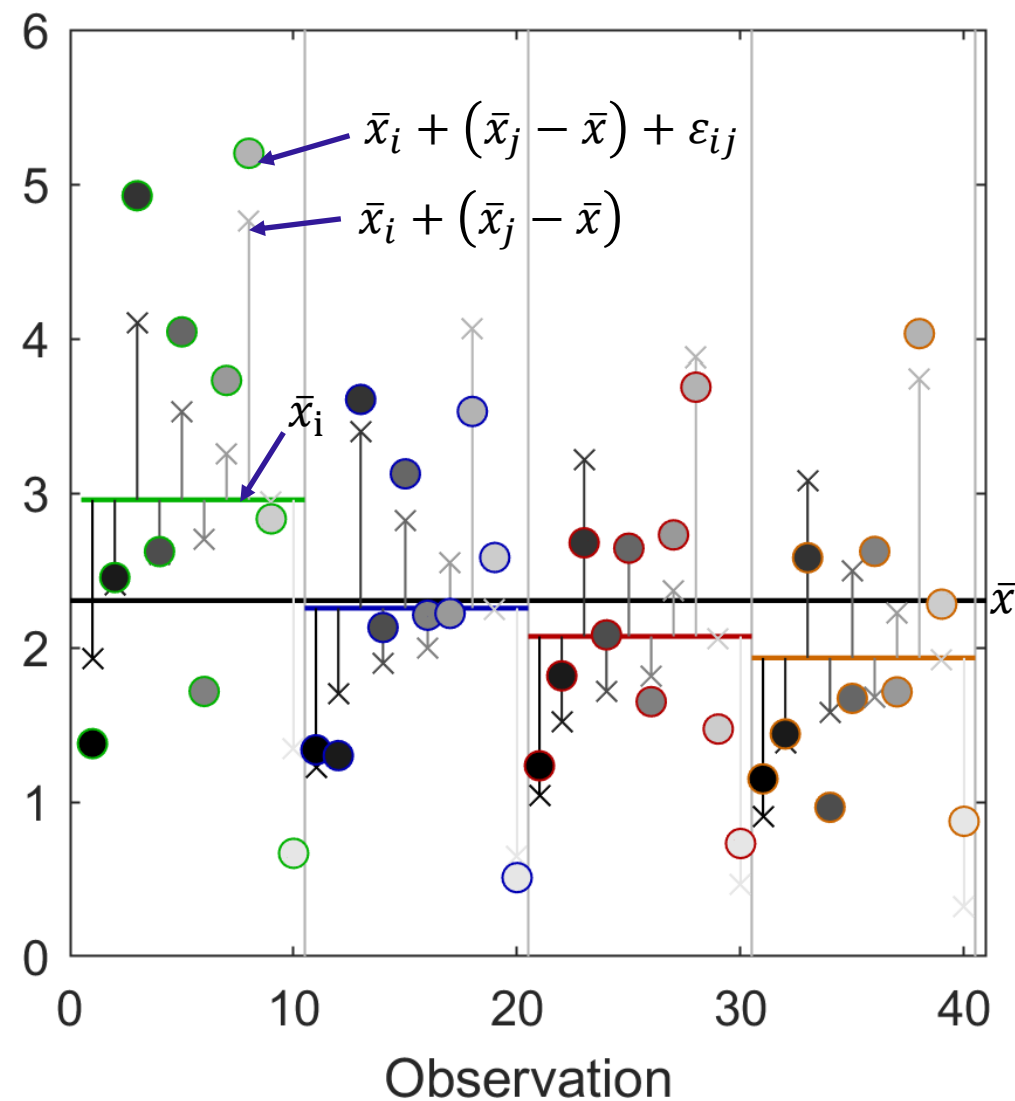
- Each group has a different group mean $\bar{X}_i$
- Each individual has their own mean
 - This yields the model (crosses)



Residual variation from the model

$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

- Each group has a different group mean $\bar{X}_i$
- Each individual has their own mean
 - This yields the model (crosses)
- Now visualise individual data points (circles)

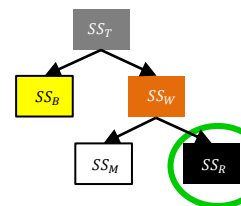
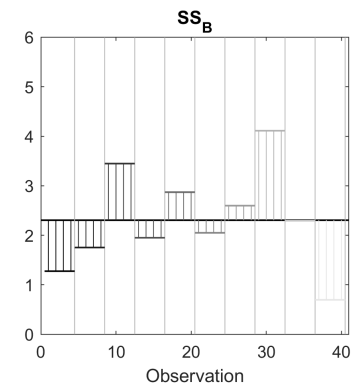
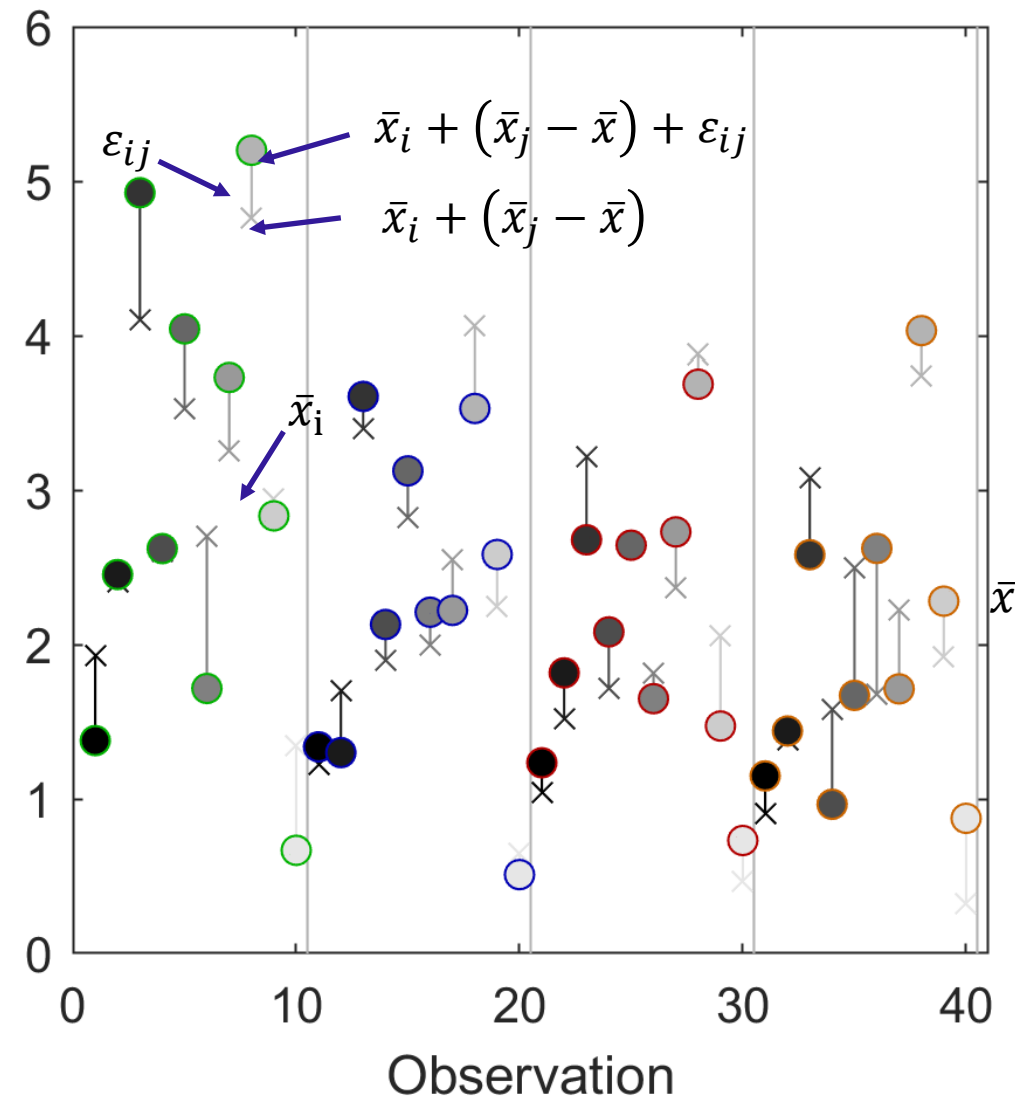


Residual variation from the model



$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

- Each group has a different group mean $\bar{X}_i$
- Each individual has their own mean
 - This yields the model (crosses)
- Now visualise individual data points (circles)
- Finally, look at difference between data and the model
- These are the residuals from the model that are not explained by the between-subjects variability





$$X_{ij} - \bar{X} = \bar{D}_i + \bar{K}_j + \varepsilon_{ij}$$

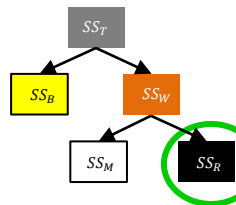
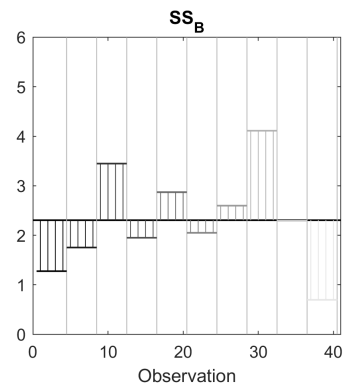
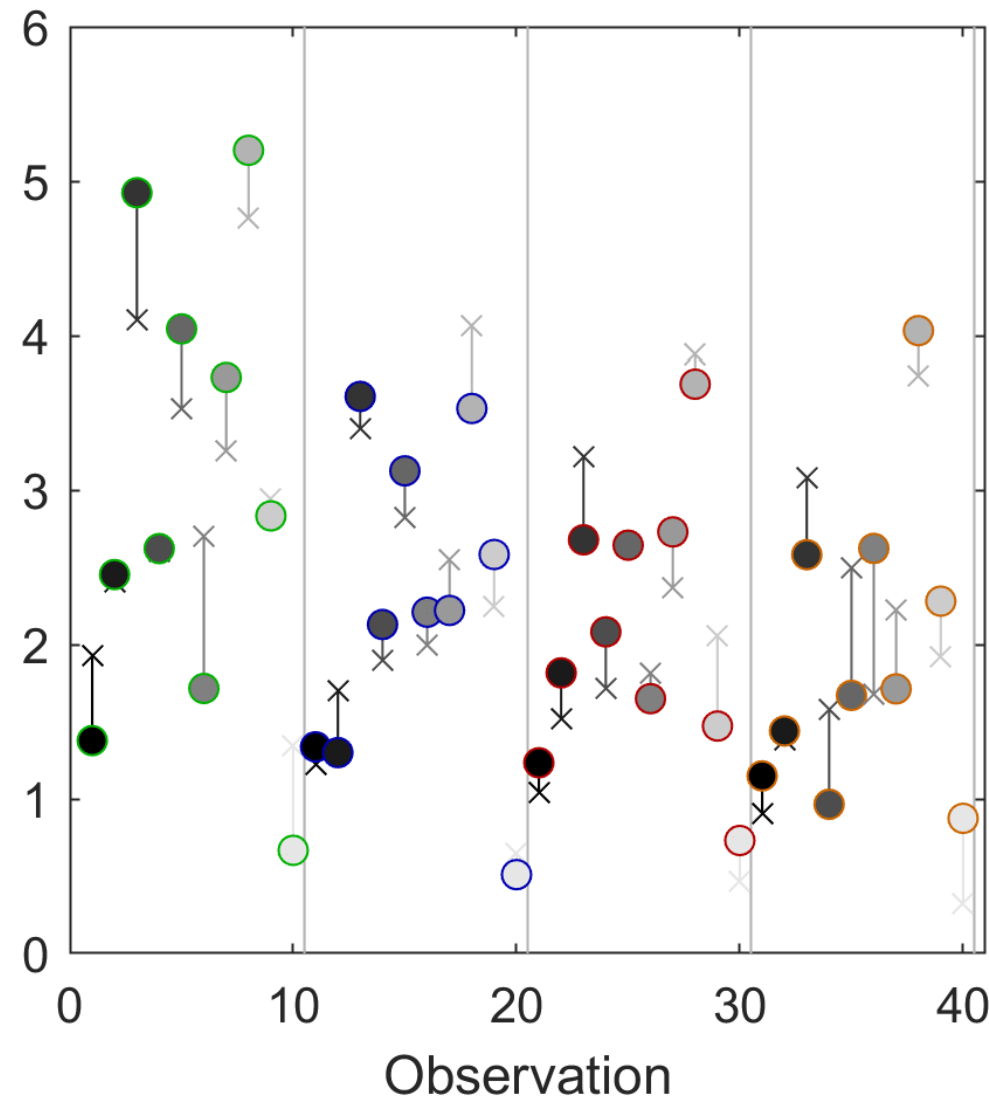
Residual Sum of Squares

Deviations from the individual values from the model (ε_{ij})

$$SS_R = \sum_{i=1}^I \sum_{j=1}^J (X_{i,j} - (\bar{X}_j + \bar{X}_i - \bar{X}))^2$$

$$\begin{aligned} \nu_R &= \nu_W - \nu_M \\ \nu_R &= N - J - (I - 1) \end{aligned}$$

$$\begin{aligned} \varepsilon_{ij} &= X_{ij} - \bar{X} - \bar{D}_i - \bar{K}_j, \\ \bar{D}_i &= \bar{X}_i - \bar{X}, \bar{K}_j = \bar{X}_j - \bar{X} \end{aligned}$$





$$MS_M = \frac{SS_M}{v_M} = \frac{SS_M}{I - 1}$$

$$MS_R = \frac{SS_R}{v_R} = \frac{SS_R}{N - J - (I - 1)}$$

$$F = \frac{MS_M}{MS_R}$$

Independent

$$MS_M = \frac{SS_M}{v_M} = \frac{SS_M}{I - 1}$$

$$MS_R = \frac{SS_R}{v_R} = \frac{SS_R}{N - I}$$

Repeated measures

$$SS_M = 6.198$$

$$SS_R = 8.264$$

Independent

$$SS_M = 6.198$$

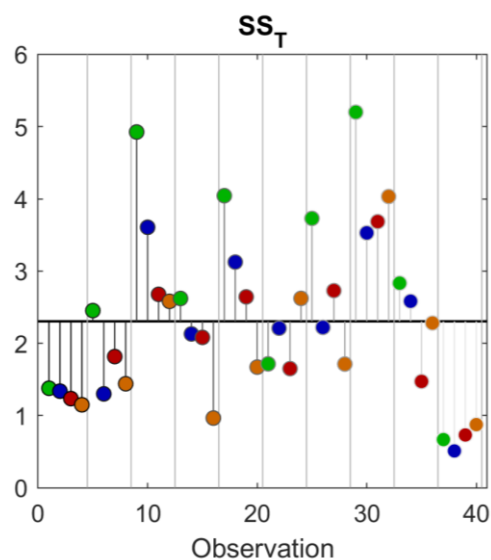
$$SS_R = 44.76$$

- If we hadn't attributed some of the variability to between-subjects, we would have attributed it to residuals, and over-estimated p
- We don't actually use SS_T , SS_B or SS_W but these are given for completeness

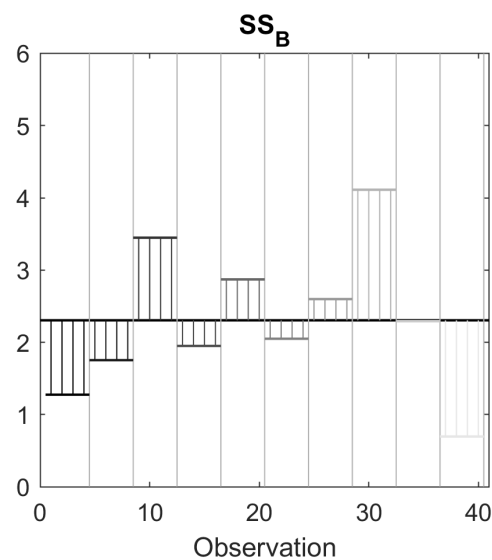


Ordered by individual, j

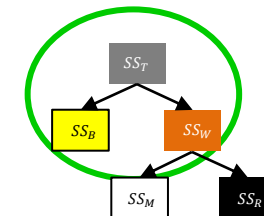
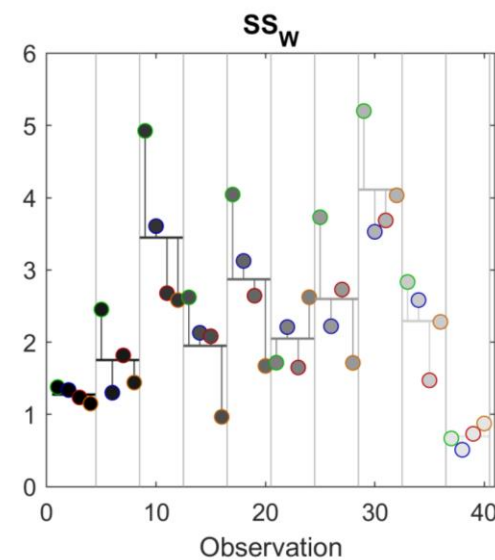
Partitioning Variability



=



+



Total variability

Explains all deviation from the grand mean

'Between' variability

Some of the variability is explained by the fact that each participant has their own baseline value

'Within' variability

Deviations from each participants own mean.

What proportion of the remaining deviations explained by the model (M) vs residuals (R)?



FYI - Total variation from the grand mean

Exactly the same as for independent

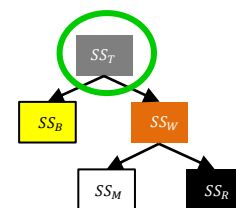
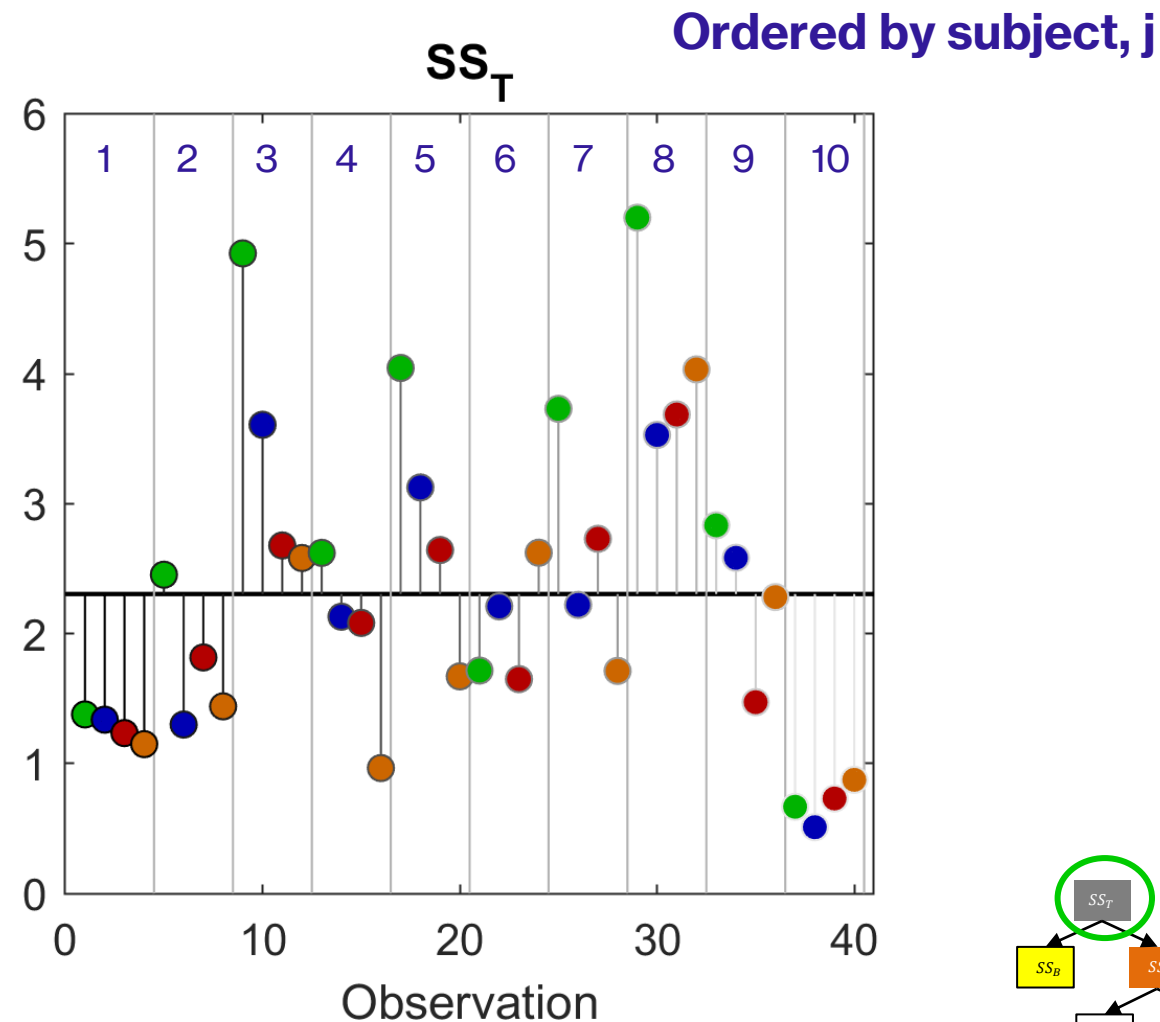
Total Sum of Squares

Deviations the individual values from the grand mean

$$SS_T = \sum_{i=1}^I \sum_{j=1}^J (X_{i,j} - \bar{X})^2$$

$$\nu_T = N - 1$$

$$SS_T = (N - 1)s_{grand}^2$$





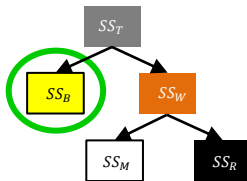
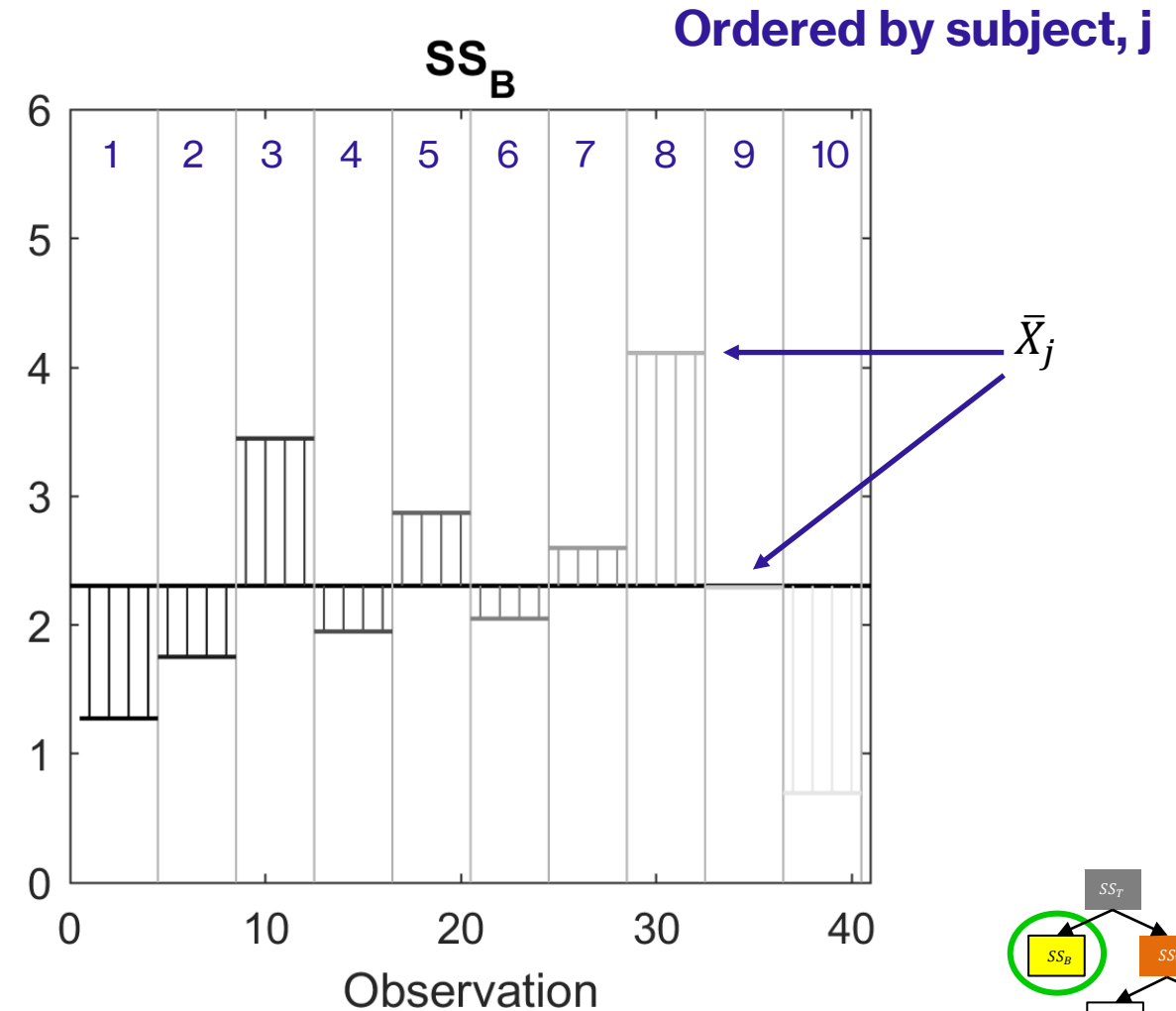
Between Subjects Sum of Squares

Deviations of the subject means from the grand mean

$$SS_B = I \sum_{i=1}^J (\bar{X}_j - \bar{X})^2$$

$$v_B = J - 1$$

This can be used to explain some of the total variability





FYI - Within Subjects Variation

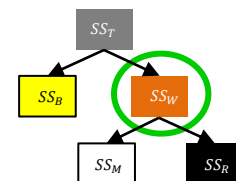
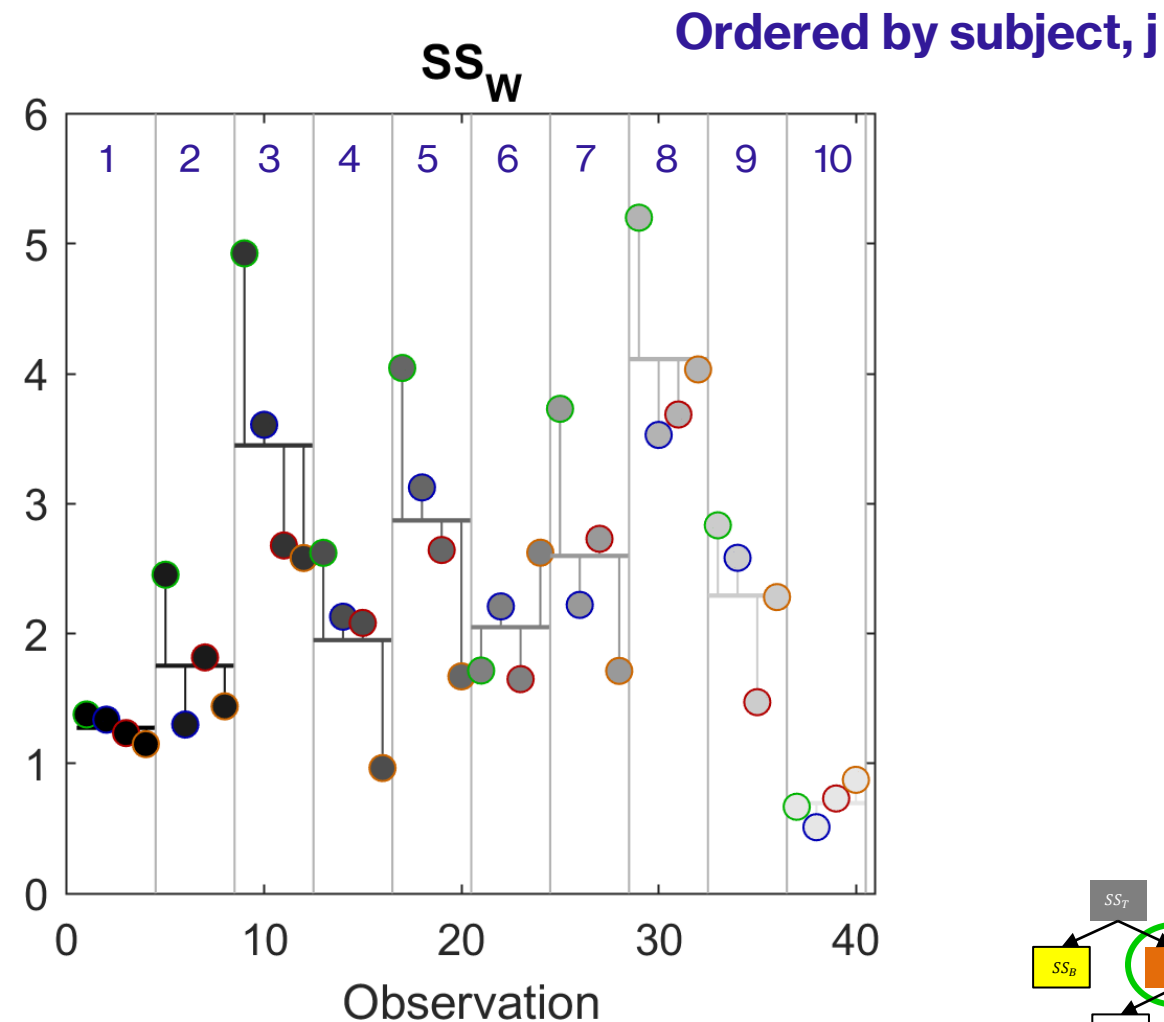
Within Subjects Sum of Squares

Deviations of the individual values from the subject means

$$SS_W = \sum_{i=1}^I \sum_{j=1}^J (X_{i,j} - \bar{X}_j)^2$$

$$v_W = J(I - 1) = N - J$$

This is the variability remaining after accounting for individual differences



Repeated Measures Notes

- For the independent ('between-group') ANOVA, we needed the variances of all groups to be similar
- For a repeated measures ANOVA ('between-subjects'), we have slightly different requirement, 'Sphericity'
- Sphericity requires similar variances of differences between groups for each subject
 - E.g. calculate difference of diamide minus healthy for each subject
 - Calculate the variance
 - This should be similar to the variances of glutaraldehyde minus healthy, glutaraldehyde minus diamide etc.
- We can use Mauchly's test to evaluate this `mauchly`
- See MATLAB with ANOVA bonus material for more detail
- There are 'corrections' that can be made if sphericity is violated to improve the test power.
- Post-hoc
 - Use Bonferroni if sphericity is violated
 - Use Tukey's test if it is not
- If normality is violated, 'Friedman's Test' is the Repeated Measures Kruskal-Wallis test (not covered in this course)





Two-way ANOVA

More than 2 groups, as defined by two characteristics

Multi-way or factorial ANOVA



- More than two groups
 - Does **Age** affect the outcomes of our rabbit experiment on vascular modelling with different **Treatments**?
- The aim is to infer whether the factors affect the population means between $G \times I$ groups
 - G the number of levels for Factor A (groups)
 - I the number of levels for Factor B (groups)
 - J in this course, balanced for simplicity
 - $N = GIJ$ the total number of observations
- For 2-way, we are comparing $G \times I$ means, $\bar{X}_{gi}$
- $\bar{X}_i$ are means for a given value of i across all g
- $\bar{X}_g$ are means for a given value of g across all i

This is my own notation, intended for consistency for 1-way and 2-way ANOVA

	Young $g = 1$			Old $g = 2$		
	Control $i = 1$	SM $i = 2$	EC $i = 3$	Control $i = 1$	SM $i = 2$	EC $i = 3$
$j = 1$	3.02	4.43	2.05	2.53	2.89	1.97
$j = 2$	3.53	4.11	2.70	1.88	2.57	1.22
$j = 3$	1.90	2.46	2.66	2.23	2.89	1.99
$j = 4$	3.14	4.21	2.67	2.54	2.52	2.35
$\bar{X}_{gi}$	2.90	3.80	2.52	2.29	2.72	1.88

	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

Null-hypotheses



- We have three null-hypotheses

1. The means of observations group by age (columns) are the same $\bar{X}_g$

- Does age have an effect, irrespective of treatment?

2. The means of observations grouped by treatment (rows) are the same, $\bar{X}_i$

- Does treatment have an effect, irrespective of age?

3. There is no interaction between the factors

- Does age affect the response to treatment?

$$\bar{X}_i = \bar{X} + \bar{D}_i$$

$$\bar{X}_g = \bar{X} + \bar{L}_g$$

	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

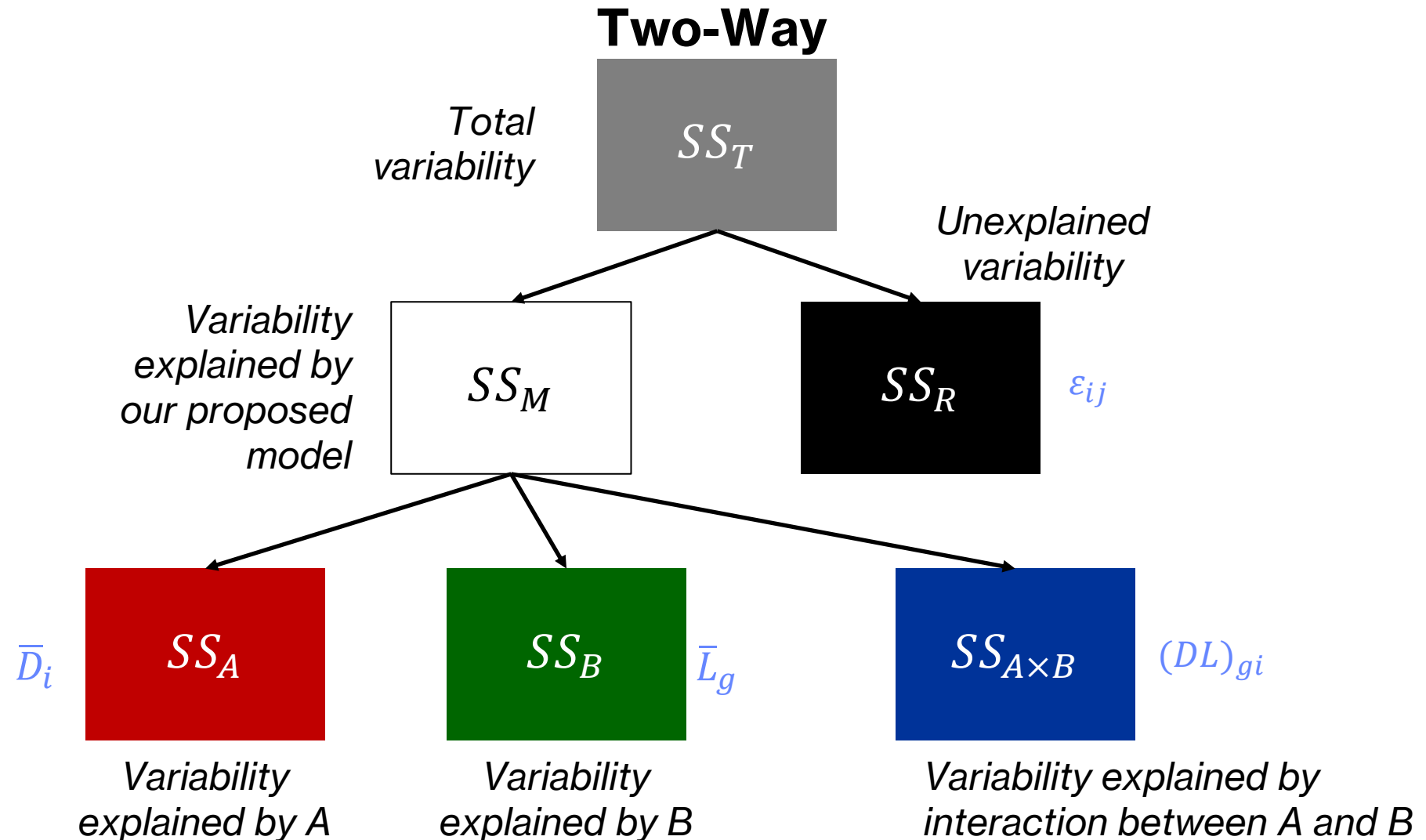
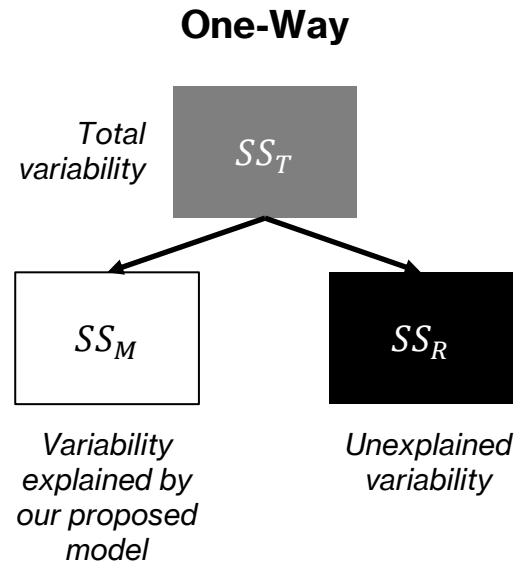
$$X_{gij} = \mu + \delta_i + \lambda_g + (\delta\lambda)_{gi} + \varepsilon_{gij}$$

- δ_i is the 'treatment effect' for rows
- λ_g is the 'treatment effect' for columns
- $(\delta\lambda)_{gi}$ is the interaction between the groups
- $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ random deviations from our model

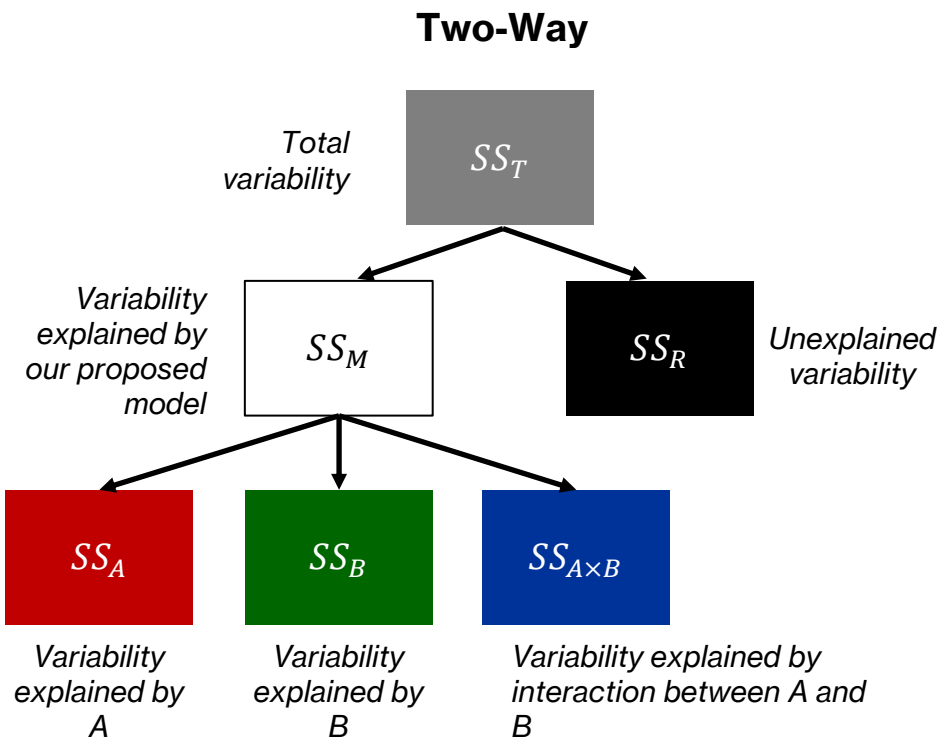
$$X_{gij} = \bar{X} + \bar{D}_i + \bar{L}_g + (DL)_{gi} + \varepsilon_{gij}$$

Partitioning Variability

$$X_{gij} - \bar{X} = \bar{D}_i + \bar{L}_g + (DL)_{gi} + \varepsilon_{gij}$$



2-way ANOVA SS_R



$G = 2$ levels of factor A (age)
 $I = 3$ levels of factor B (treatment)
 $J = 4$ observations per group
 $N = 24$ total observations

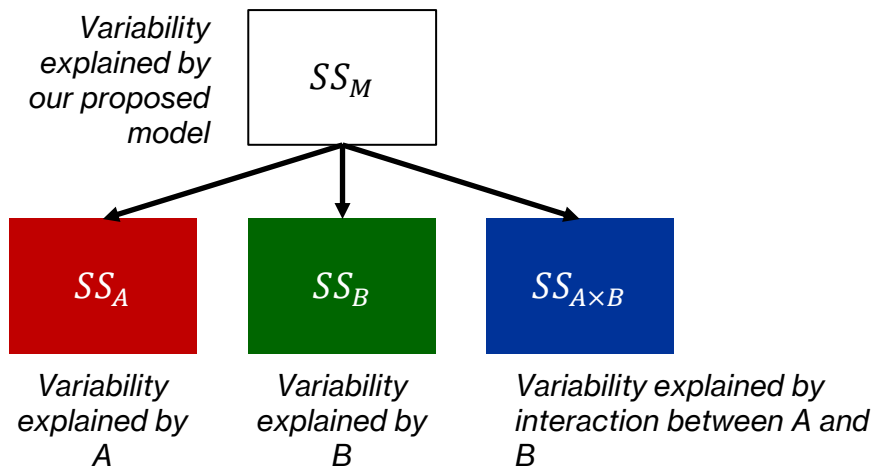
	Young $g = 1$			Old $g = 2$		
	Control $i = 1$	SM $i = 2$	EC $i = 3$	Control $i = 1$	SM $i = 2$	EC $i = 3$
$j = 1$	3.02	4.43	2.05	2.53	2.89	1.97
$j = 2$	3.53	4.11	2.70	1.88	2.57	1.22
$j = 3$	1.90	2.46	2.66	2.23	2.89	1.99
$j = 4$	3.14	4.21	2.67	2.54	2.52	2.35
$\bar{X}_{gi}$	2.90	3.80	2.52	2.29	2.72	1.88

$$SS_R = \sum_{g=1}^G \sum_{i=1}^I \sum_{j=1}^J (X_{gij} - \bar{X}_{gi})^2$$

$$v_R = N - GI$$

SS of each column in table above minus its column mean

2-way ANOVA $SS_A - SS_M$ grouped by G



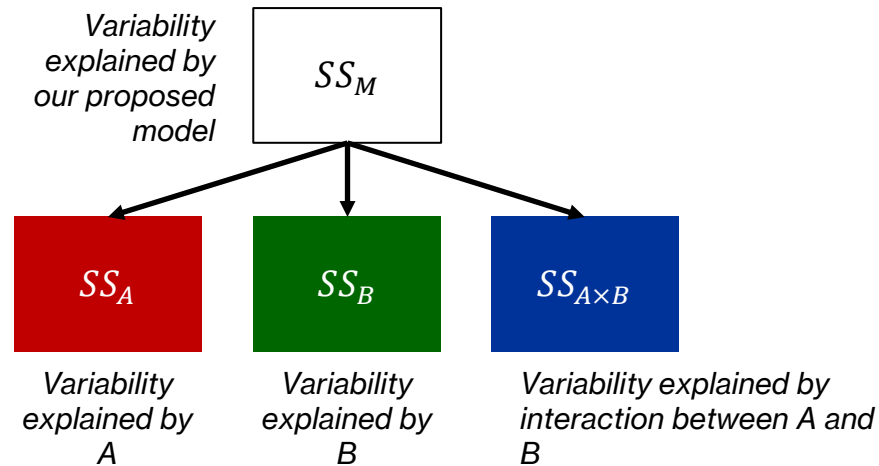
$$SS_A = IJ \sum_{g=1}^G (\bar{X}_g - \bar{X})^2$$
$$v_A = G - 1$$

	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

SS of all individual group means after grouping by G, relative to the grand mean

Model: age can predict outcome, irrespective of treatment

2-way ANOVA $SS_B - SS_M$ grouped by I



	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

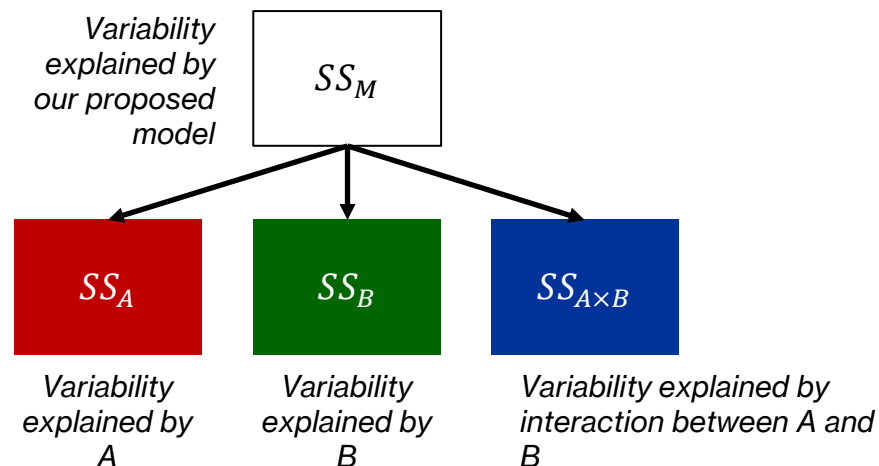
$$SS_B = GJ \sum_{i=1}^I (\bar{X}_i - \bar{X})^2$$

$$v_B = I - 1$$

SS of all individual group means after grouping by I, relative to the grand mean

Model: treatment can predict outcome, irrespective of age

2-way ANOVA $SS_{A \times B}$ – left over SS



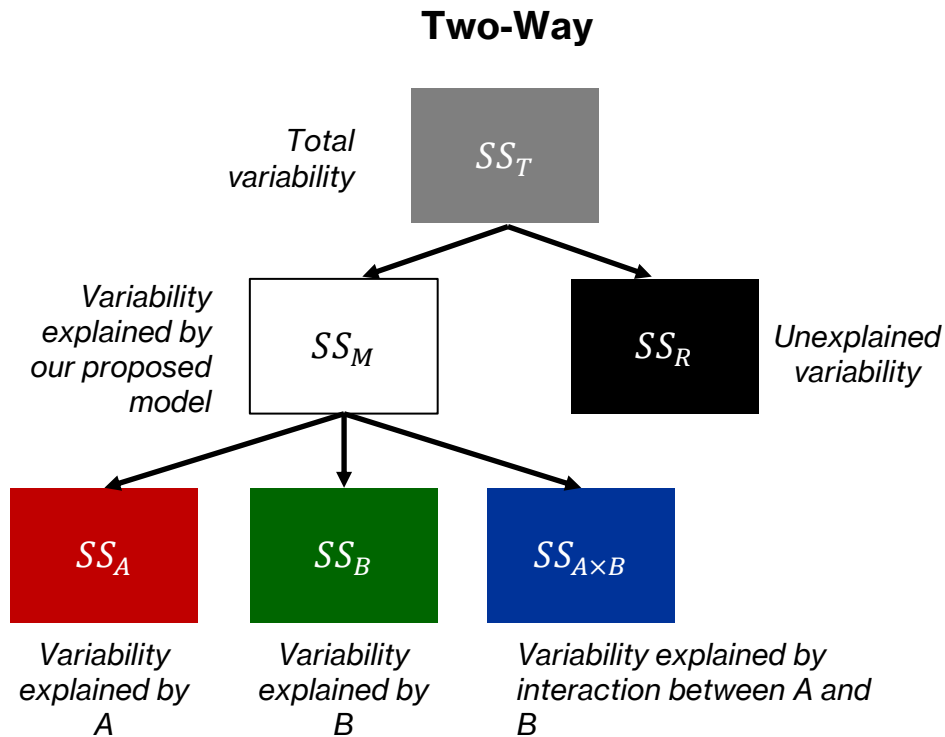
$$SS_{A \times B} = SS_M - SS_A - SS_B$$

$$v_{A \times B} = v_M - v_A - v_B$$

Residual SS not accounted for by factors A or B

Must be attributed to correlations between the factors

2-way ANOVA MS and F



$$MS_A = \frac{SS_A}{v_A}$$

$$MS_B = \frac{SS_B}{v_B}$$

$$MS_{A \times B} = \frac{SS_{A \times B}}{v_{A \times B}}$$

$$MS_R = \frac{SS_R}{v_R}$$

$$F_A = \frac{MS_A}{MS_R}$$

$$F_B = \frac{MS_B}{MS_R}$$

$$F_{A \times B} = \frac{MS_{A \times B}}{MS_R}$$

On reference sheet

2-way ANOVA table

Hypotheses

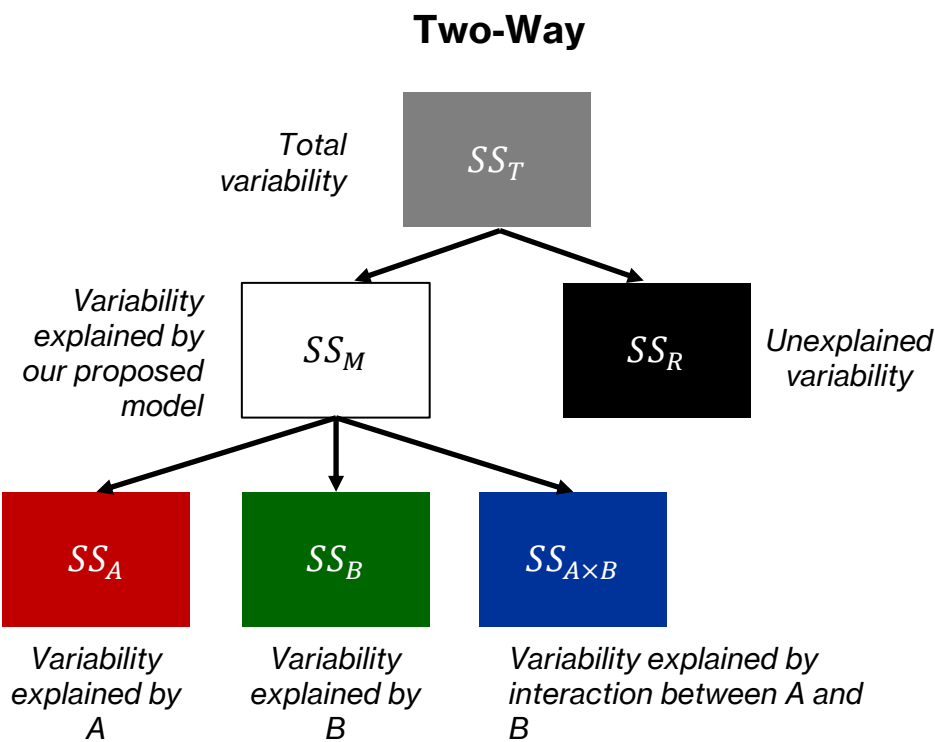
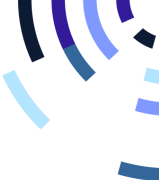
We have three null-hypotheses

1. The means of observations grouped by treatment (rows) are the same, $\bar{X}_i$
2. The means of observations group by age (columns) are the same $\bar{X}_g$
3. There is no interaction between the factors (age does not affect the response to treatment)

ANOVA Table					
Source	SS	df	MS	F	Prob>F
Columns	3.618	1	3.61796	12.22	0.0026
Rows	4.593	2	2.2965	7.76	0.0037
Interaction	0.2885	2	0.14426	0.49	0.6221
Error	5.3276	18	0.29598		
Total	13.8271	23			

	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

FYI - 2-way ANOVA SS_T



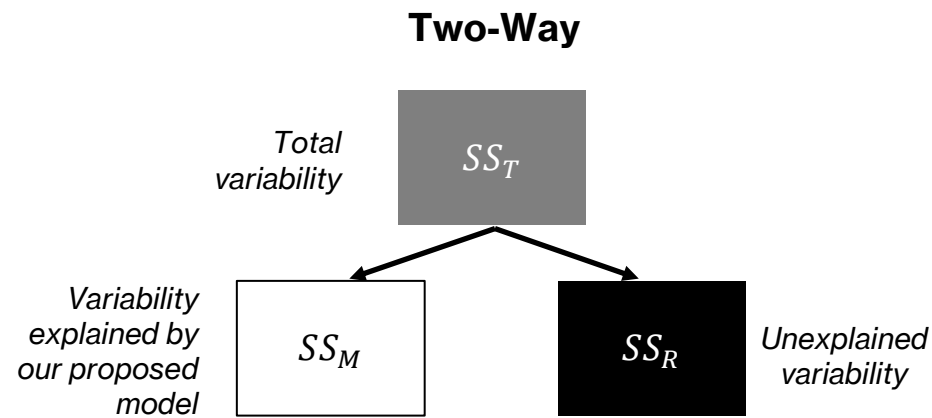
ANOVA Table					
Source	SS	df	MS	F	Prob>F
Columns	3.618	1	3.61796	12.22	0.0026
Rows	4.593	2	2.2965	7.76	0.0037
Interaction	0.2885	2	0.14426	0.49	0.6221
Error	5.3276	18	0.29598		
Total	13.8271	23			

$$SS_T = \sum_{g=1}^G \sum_{i=1}^I \sum_{j=1}^J (X_{gij} - \bar{X})^2$$
$$v_T = N - 1$$

SS of all individual values
minus the grand mean

- $G = 2$ levels of factor A (age)
- $I = 3$ levels of factor B (treatment)
- $J = 4$ observations per group
- $N = 24$ total observations

FYI - 2-way ANOVA SS_M



$\bar{X}_{gi}$	Young $g = 1$	Old $g = 2$	$\bar{X}_i$
Control $i = 1$	2.90	2.29	2.59
SM $i = 2$	3.80	2.72	3.26
EC $i = 3$	2.52	1.88	2.20
$\bar{X}_g$	3.07	2.30	

$$SS_M = \sum_{g=1}^G \sum_{i=1}^I J(\bar{X}_{gi} - \bar{X})^2$$

$$\nu_M = GI - 1$$

SS of all individual group means relative to the grand mean

Carrying out 2-way ANOVA



- Assume you're given the MS values and use $\alpha = 0.05$

$$F_A = \frac{MS_A}{MS_R}$$

$$F_B = \frac{MS_B}{MS_R}$$

$$F_{A \times B} = \frac{MS_{A \times B}}{MS_R}$$

ANOVA Table					
Source	SS	df	MS	F	Prob>F
Columns	3.618	1	3.61796	12.22	0.0026
Rows	4.593	2	2.2965	7.76	0.0037
Interaction	0.2885	2	0.14426	0.49	0.6221
Error	5.3276	18	0.29598		
Total	13.8271	23			

- We can conclude that
 - factor A (age) has an effect on the average carotid diameter
 - factor B (treatment) has an effect on the average carotid diameter
- There is not a statistically detectable interaction

		v ₁									
		1	2	3	4	5	6	7	8	9	10
v ₂	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.43	2.38
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.40	2.35

What if there were interactions?



- A statistical interaction tells us that there is an interaction between age and treatment on the outcome
- I.e. there is a differential response to the EC or SM treatment in young or older animals
- Factor A affects the response to Factor B
- The factors are not independent
- Hence any effects analysed by grouping for a single factor are not valid (as the results depend on the choice of the other factor)
- **If there is an interaction, do not further analyse the other hypotheses**

Different simulated data set

ANOVA Table					
Source	SS	df	MS	F	Prob>F
Columns	0.7577	1	0.75765	4.1	0.058
Rows	5.4528	2	2.72639	14.75	0.0002
Interaction	2.1301	2	1.06504	5.76	0.0116
Error	3.3279	18	0.18488		
Total	11.6684	23			

- Can be useful if you want to find out whether there is an interaction!
- If no interactions, proceed with post-hoc tests
 - One for each factor

Lecture 9: Multiple Comparisons

What we covered

- Introduction to ANOVA
 - Why not multiple t-tests?
 - Bonferroni and Bonferroni-Holm
 - Sums of Squares
 - Partitioning Variability
 - The F-statistic
- Post hoc tests
- Kruskal-Wallis Test
- Repeated Measures ANOVA
 - Partitioning Variability
- Two-Way ANOVA
 - Partitioning Variability
 - Interactions

Up next
Final Reflections



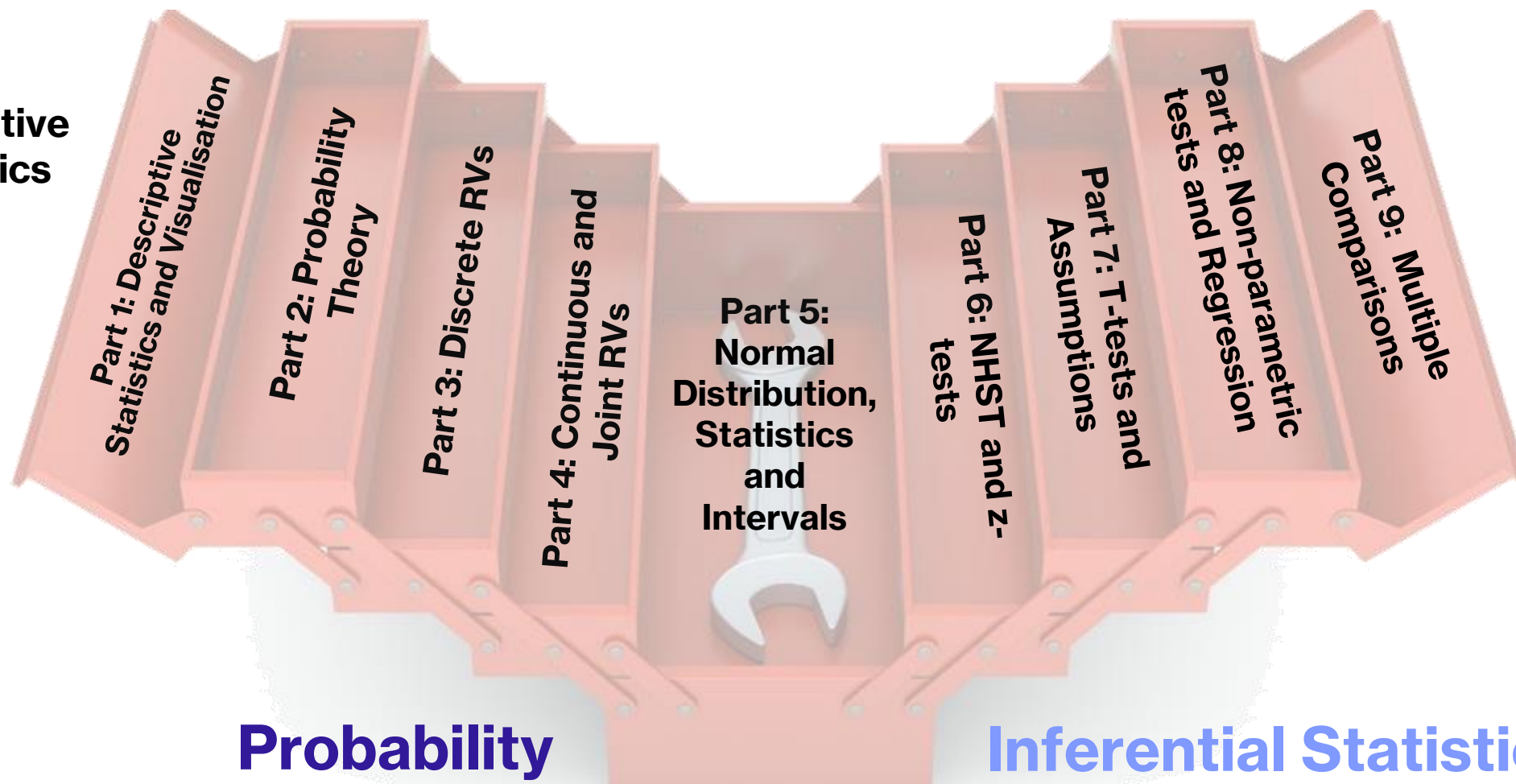
A final test

Essential, but not sufficient

The 'Swear on Your ...' test

- There is one final statistical test that is more important than all the others...
 - Does not have a null hypothesis
 - **Is essential but not sufficient**
 - Cannot be used in place of more formal statistics
- It is derived on the basis that if you acquired and analysed the data, then you know it better than anyone else
- It is a thought experiment
 - If you had to swear on your [insert appropriate important thing here]
 - Do you really trust that the conclusions you are making are justified?
 - Sometimes called the 'gun-to-your-head' test or 'trust-your-gut' test
- If the answer is no, then revisit your analysis
- Assumptions:
 - The pressure to 'get a result' is not clouding your judgement
 - You have a basic understanding of statistics, such as gained from this course

What we covered

Descriptive
Statistics

Farewell Limerick

**A student of bioengineering,
had a dislike for stats that was searing.
But after the lecture,
was inspired to conjecture,
that the subject can be quite endearing.**

